

JAX-based codes for WDM-transform for use within gravitational wave astronomy

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The WDM transform has been widely used in gravitational-wave astronomy, yet a self-contained, mathematically explicit reference for practitioners remains lacking. This is especially true for those wishing to adopt the transform in modern Python and JAX inference workflows. We present `wdm_transform`, an open-source Python package implementing the Wilson–Daubechies–Meyer (WDM) wavelet-packet time-frequency transform for gravitational-wave data analysis. We document the forward and inverse transform step by step, making explicit the packed $(N_t, N_f + 1)$ coefficient layout, the DC and Nyquist edge-channel conventions, the alternating phase factor C_{nm} , and the normalization conditions required for exact round-trip reconstruction. The package supplies NumPy and JAX backends, both round-trip validated to floating-point precision, with the JAX backend enabling GPU-accelerated transforms of million-point data streams in tens of milliseconds. As a worked example we verify that the WDM-domain Whittle likelihood reproduces frequency-domain posteriors for a resolved LISA galactic binary under a shared stationary noise model, confirming numerical equivalence of the two representations in that controlled setting.

I. INTRODUCTION

Time-frequency analysis is widely used in gravitational-wave astronomy. It enables template-free characterisation of source signals [18, 26], instrumental transients [24], and the joint analysis of astrophysical signals and terrestrial disturbances [8]. Time-frequency representations also provide compact waveform descriptions and facilitate the modelling of non-stationary detector noise. Among the available techniques, discrete wavelet-packet transforms are particularly well suited to gravitational-wave data analysis: they furnish localized, orthonormal basis functions capable of representing transient and chirping signals, and for locally stationary noise they yield an approximately diagonal noise covariance, greatly simplifying likelihood evaluation and spectral estimation.

The Wilson–Daubechies–Meyer (WDM) wavelet-packet basis already has a substantial history in gravitational-wave astronomy. The fast Wilson–Daubechies transform was developed for transient gravitational-wave analysis by Nuccia, Klimenko, and Mitselmakher [21], and WDM representations have been used in coherent WaveBurst and related burst-search contexts [15, 25]. These earlier papers and implementations established the theoretical and algorithmic basis of the transform for excess-power and time-frequency analyses, including its orthonormal structure, controlled

spectral leakage, flexible frequency sub-bands, and time-delay filters relevant for sky localization. Subsequent WDM software and applications by Digman and Cornish further developed fast forward and inverse transforms, WDM likelihoods, and time-frequency analyses for LISA sources and backgrounds [12–14]. Recent work by Pearson and Cornish has also shown how WDM-domain methods can be combined with Bayesian data augmentation to handle data gaps in long-duration gravitational-wave observations [22].

Cornish developed a complementary WDM-domain programme for likelihood-based gravitational-wave inference and waveform generation. In this formulation, regularly sampled data are mapped to an orthogonal grid of localized time-frequency pixels; slowly evolving signals may be represented compactly as tracks on this grid, and binary waveforms can be generated directly in the WDM domain [5]. A central statistical motivation is that locally stationary noise, whose spectrum varies slowly across an individual time-frequency pixel, has an approximately diagonal covariance in the WDM domain. This approximate diagonalization, analyzed in detail in Ref. [7], provides the WDM-domain analogue of the familiar diagonal Fourier-domain covariance for stationary noise and underlies WDM Whittle-type likelihoods. A clear one-to-one correspondence between equations and their implementation is often missing in existing literature and codebases [6, 9, 11, 20]: We ad-

dress this gap here.

The present manuscript does not claim to introduce the WDM transform, to replace these foundational analyses, or to outperform existing pipeline-specific implementations. Its contribution is instead to make the sampled WDM construction easy to adopt and reproduce in modern Python-based gravitational-wave workflows. We give a self-contained derivation of the forward and inverse transforms, state the packed $(N_t, N_f + 1)$ coefficient layout explicitly, spell out the DC and Nyquist channel treatment, and fix the normalization, phase, and indexing conventions used by the implementation. These choices are then validated through exact round-trip reconstruction tests and empirical checks of coefficient decorrelation. The accompanying `wdm_transform` package provides CPU and GPU-capable NumPy/JAX backends, so that the same mathematically explicit transform can be used in conventional analyses and in differentiable inference workflows.

The motivation for the WDM can be understood by contrasting the two standard representations of a discrete signal. In the time domain one can identify when a feature occurs but not which frequency band is active, whereas in the Fourier domain one can identify which frequencies are present but not when they appear. Many gravitational-wave signals and detector artefacts are neither purely stationary nor purely impulsive, so neither representation alone is fully satisfactory. A short-time Fourier often makes use of compactly-supported sliding windows such as square or Tukey windows. In contrast, WDM employs frequency-localized atoms whose corresponding time-domain representation exhibits algebraically decaying tails proportional to $1/t^3$.

The result is a uniform time-frequency representation well suited to describing transients, chirps, gaps, and other localized structure, while still admitting an exact inverse transform back to the original sampled data.

The remainder of this paper is organized as follows. Section [IA](#) introduces notation and conventions. Section [II](#) develops the WDM transform, including the Meyer window, the full three-case basis definition covering DC, interior, and Nyquist channels, the forward and inverse transforms, and the statistical properties of the coefficients under stationary and non-stationary noise. Section [IV](#) describes the `wdm_transform` package, presents runtime and accuracy benchmarks, and demonstrates the transform on astrophysical signals. We conclude in Section [V](#).

A. Conventions

Consider a data stream consisting of N uniformly sampled values of a **real** quantity x at times

$\{t_n\}_{n=0,\dots,N-1}$ over a total observation time T :

$$x[n] = x(t_n) \quad , \quad n = 0, \dots, N-1 \quad , \quad (1)$$

with sampling interval $\Delta t = t_{k+1} - t_k$ and $T = N\Delta t$. All available information about the signal $x[n]$ is encoded in these N samples. We define the sampling rate and the Nyquist frequency as

$$\begin{aligned} F_s &\equiv \frac{1}{\Delta t} \equiv \frac{N}{T} \quad , \\ f_{\max} &= \frac{1}{2} F_s \quad , \\ \Delta f &= 1/T \end{aligned} \quad (2)$$

In that regard, if we set $t_0 = 0$, we can write $t_n = n\Delta t$. In the frequency domain, the data stream $\{x_n\}_{n=0}^{N-1}$ can be discrete-Fourier-transformed (DFT) to a set of other N complex numbers as

$$\tilde{x}[l] = \sum_{k=0}^{N-1} x[k] e^{-2\pi i l k / N} \quad (3)$$

for Fourier frequencies $f_l = \{-N\Delta f/2, \dots, -\Delta f, 0, \Delta f, \dots, (N/2 - 1)\Delta f\}$, i.e. the zero-frequency bin is centred in the array. Bold faced quantities $\mathbf{A}$ stand for vector/matrix quantities with components A_{ij} . Dagger based operations $\mathbf{A}^\dagger$ stand for the conjugate transpose of the matrix $\mathbf{A}$.

We assume that the output of a gravitational wave detector, $x(t)$, is expressed as a linear combination of a signal, $h(t; \boldsymbol{\theta})$, determined by a finite set of (unknown) parameters, $\boldsymbol{\theta}$, and instrumental noise, $n(t)$. If we ignore the presence of calibration errors [\[?\]](#), the content of a single data stream, i.e., one output channel from one detector, can be written in the time and frequency domain as:

$$x[k] = h[k; \boldsymbol{\theta}] + n[k] \quad , \quad (4)$$

$$\tilde{x}[l] = \tilde{h}[l; \boldsymbol{\theta}] + \tilde{n}[l] \quad (5)$$

where the tilde indicates the Fourier transform.

II. THE WILSON–DAUBECHIES–MEYER (WDM) TRANSFORM

A. Terminology

The name Wilson–Daubechies–Meyer combines several pieces of time-frequency analysis terminology. A standard discrete wavelet transform usually refers to a multiresolution, dyadic decomposition (“dyadic” meaning organized by factors of two, so that each successive scale changes the time and frequency resolution

by powers of two). Such transforms represent low-frequency structure with broad time support and fine frequency resolution, and high-frequency structure with narrower time support and coarser frequency resolution. A wavelet packet (or wavelet-wavepacket) transform is more general: it allows a broader class of admissible tilings of the time-frequency plane. The object studied in this paper is the Wilson–Daubechies–Meyer (WDM) wavelet-packet transform; for brevity, we refer to it as the WDM transform throughout. Unlike a standard dyadic wavelet transform, the WDM transform used here has a uniform rectangular tiling, so that all pixels have the same time width ΔT and frequency width ΔF , with $\Delta T \Delta F = 1/2$ under the one-sided frequency convention used below [10].

The naming of the transform reflects two distinct ingredients: the Wilson-basis pairing of positive- and negative-frequency atoms, introduced together with the phase factor, and the smooth Daubechies–Meyer frequency window used to construct those atoms, all of which introduced together with its closed form in Sec. II below.

The WDM transform maps the original N samples to a packed coefficient grid with N_t time bins and $N_f + 1$ stored frequency channels. The transform size is still set by

$$N = N_t N_f, \quad (6)$$

and the stored channels are indexed by $m = 0, \dots, N_f$, with $m = 0$ the DC edge channel, $m = 1, \dots, N_f - 1$ the interior channels, and $m = N_f$ the Nyquist edge channel. The underlying idea is to partition the data stream into N_t time segments and to perform a frequency domain analysis of each one. Here N is fixed by the data, while N_t is a free user choice subject only to $1 \leq N_t \leq N$ (and consequently $1 \leq N_f \leq N$). Given that choice we can define the spacing in both time and frequency with

respect to the underlying tiling with area $N = N_t N_f$

$$\begin{aligned} \Delta T &\equiv \frac{T_{\text{obs}}}{N_t} && \text{grid time spacing,} \\ \Delta F &\equiv \frac{f_{\text{max}}}{N_f} && \text{grid frequency spacing.} \end{aligned} \quad (7)$$

One immediately sees that,

$$\Delta T = \frac{T_{\text{obs}}}{N_t} = \frac{N \Delta t}{N_t} = N_f \Delta t \quad (8)$$

$$\Delta F = \frac{f_{\text{max}}}{N_f} = \frac{1}{2 N_f \Delta t} = \frac{N_t \Delta f}{2} = \frac{1}{2 \Delta T} \quad (9)$$

This final expression shows how the Nyquist theorem still holds. Once you fix N_t (or equivalently N_f , ΔT or ΔF) the grid is uniquely identified. Figure 1 summarizes the packed layout, including the DC and Nyquist edge channels and the alternating phase factor used for the interior coefficients. Note that N_t and N_f are required to be even — they need not be powers of two. Relaxing the evenness requirement is left to future work.

B. Forward transform

For a given frequency series $\tilde{x}[l]$, the WDM transform represents the data through the WDM coefficients w_{nm}

$$w_{nm} = \sum_{l=0}^{N-1} \tilde{x}[l] \tilde{g}_{nm}^*[l], \quad (10)$$

where n corresponds to the time index and m corresponds to the frequency-channel.

The basis elements $\tilde{g}$, FD window $\tilde{\varphi}$ and constants C are defined in closed form in the discrete frequency domain below. For DC ($m = 0$), interior channels ($0 < m < N_f$), and Nyquist ($m = N_f$) we have that:

$$\textbf{Wilson Basis: } \tilde{g}_{nm}[l] = \frac{1}{\sqrt{2}} \begin{cases} e^{-4\pi i n l / N_t} \tilde{\varphi}[l], & m = 0, \\ e^{-2\pi i n l / N_t} \left(C_{nm} \tilde{\varphi}[l - m N_t / 2] + C_{nm}^* \tilde{\varphi}[l + m N_t / 2] \right), & 0 < m < N_f, \\ e^{-4\pi i n l / N_t} \left(\tilde{\varphi}[l - N/2] + \tilde{\varphi}[l + N/2] \right), & m = N_f, \end{cases} \quad (11)$$

$$\textbf{Daubechies-Meyer: } \tilde{\varphi}[l] = \sqrt{\frac{2}{N_t}} \begin{cases} 1 & |l_r| < A \\ \cos \left[\frac{\pi}{2} \left(\frac{|l_r| - A}{B} \right) \right] & A \leq |l_r| < A + B, \\ 0 & \text{otherwise,} \end{cases} \quad l_r = l \frac{2}{N_t}, \quad (12)$$

$$C_{nm} = e^{\frac{i\pi}{4}(1 - (-1)^{n+m})} = \begin{cases} 1 & n + m \text{ even,} \\ i & n + m \text{ odd.} \end{cases} \quad (13)$$

$$B = 1 - 2A, \quad \text{for } A \in [0, 1/2]. \quad (14)$$

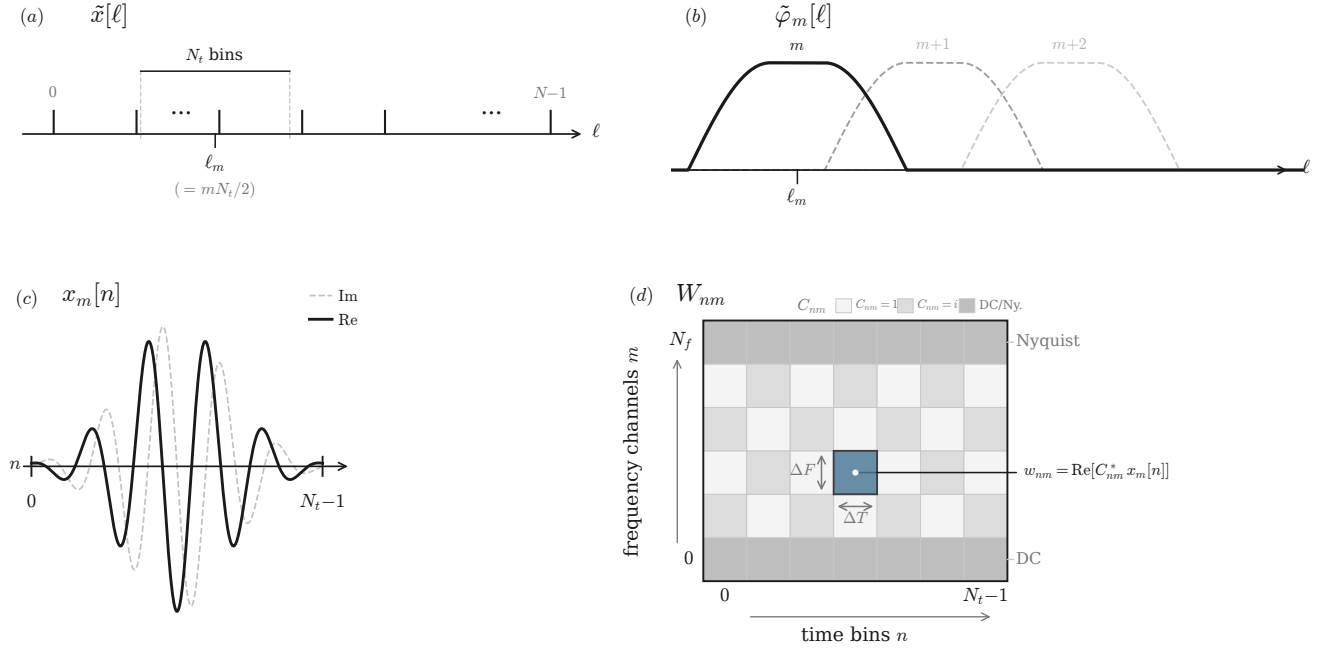


Figure 1. Schematic illustration of the WDM forward transform. (a) The input DFT spectrum $\tilde{x}[\ell]$ with channel m spanning N_t bins centred at $mN_t/2$. (b) The Meyer window $\tilde{\varphi}_m[\ell]$ shifted across N_f channels. (c) The time-domain atom $x_m[n]$ obtained by applying the inverse DFT to the windowed channel. (d) The packed coefficient array W_{nm} , with each cell $w_{nm} = \text{Re}[C_{nm}^* x_m[n]]$; the phase factor C_{nm} alternates between 1 and i on a checkerboard pattern (see legend).

where, in the codebase, we represent the dimensionless frequency arrays via $\hat{f}[l] = f[l]/\Delta F$, for ΔF the spacing between frequencies in the WDM domain. Equation (14) gives freedom to the user to define the properties of the Daubechies-Meyer window defined in Eq.(12).

From this point onwards, we will provide an interpretation for each of the equations (11) – (14). We begin by discussing the DC, Interior, Nyquist split of the basis functions $\tilde{g}$, an important feature of our codebase, and the practical and theoretical consequences of the Daubechies-Meyer window $\tilde{\varphi}$.

From Eq.(13), the interior channels carry an alternating phase factor C_{nm} , while the DC and Nyquist edge channels have a doubled time-shift exponent and no C_{nm} modulation. The phase factor C_{nm} is the “Wilson” ingredient of the WDM construction: it combines the positive- and negative-frequency windowed Fourier atoms centered at $mN_t/2$ into a single real, orthogonal pair, which is what makes the interior basis real-valued and Wilson-orthogonal, to be discussed later in this section, in the discrete setting.

These edge channels $m = 0$ and $m = N_f$ in (11) each contribute N_t real degrees of freedom to the packed grid: the interior block stores $N_t(N_f-1)$ real coefficients via the C_{nm} -modulated pairing of the $\pm mN_t/2$ atoms (the

Wilson-basis construction), while the DC and Nyquist channels each contribute N_t unpaired real coefficients. The total $N_t(N_f+1)$ stored entries therefore represent the same $N = N_tN_f$ time-domain degrees of freedom, with the edge channels overcomplete by exactly N_t real numbers [5, 21]. We refer the reader to Fig.1 (a) and (d), for a schematic of the WDM transform.

The window $\tilde{\varphi}$ is the “Daubechies-Meyer” ingredient of the WDM construction: a smooth, compactly supported frequency window of the Meyer family, whose taper region overlaps between adjacent channels in a way that yields exact partition-of-unity reconstruction. In the present implementation, and in this work, we use the $d = 1$ cosine-tapered member of this family. We do stress that the parameter d is retained in the Application Programming Interface (API) only for possible future generalizations.¹ By construction, $\tilde{\varphi}$ has a limited frequency bandwidth which is controlled by the pa-

¹ Cornish [5] uses $d = 4$, which yields a flatter passband and steeper roll-off at the expense of a wider time-domain support (the window function must be truncated at $\pm q\Delta T$ with $q = 16$). The $d = 1$ cosine taper used here has a gentler roll-off but is more compact in time, requiring a smaller q . In practice, the choice of d trades spectral leakage suppression against temporal compactness of the window; for the applications considered

parameter $A \in [0, 1/2]$ in Eq.(14). The Daubechies-Meyer window in the frequency domain is given by Fig.1 (b). At each wavelet frequency bin l , one sees that the window overlaps with adjacent frequency bins.

For our implementation we use the default $A = 1/3$ and hence $B = 1/3$, so the window is non-zero for the normalised frequencies $|\hat{f}| \leq 2/3$. This is a default, not a hard-coded constraint: A is exposed in

`wdm_transform` as the parameter `a` of the `WDM` class (with $B = 1 - 2A$ derived automatically), and any choice with $A \in (0, 1/2)$ is admissible.

A final and essential (!) property of the WDM formalism is the principle of orthogonality. The wilson basis $\tilde{g}$ defined by Eq. (11) combined with the Debauchies-Meyer window Eq.(12) with associated constants (13) provide an orthogonal basis²:

$$\begin{aligned} \sum_{l=-N/2}^{N/2-1} \tilde{g}_{nm}^*[l] \tilde{g}_{pq}[l] &= \delta_{m,q} \begin{cases} \frac{1}{2} (\delta_{n,p} + \delta_{n,p \pm \frac{N_t}{2}}) & m \in \{0, N_f\} \\ \delta_{n,p} & 0 < m < N_f \end{cases} \\ \sum_{n=0}^{N_t-1} \sum_{m=0}^{N_f} \tilde{g}_{nm}^*[l] \tilde{g}_{nm}[l'] &= \delta_{ll'} , \end{aligned} \quad (15)$$

Both equalities are proved in in appendix A using Eq.(??) and Eq.(13). The basis functions here are represented in the frequency domain and it is easily proved they are orthogonal in the time domain. Orthogonality is the one of the key reasons that the WDM formalism is adept for gravitational wave data analysis, which will be discussed at a later stage of this work.

These are the main ingredients behind the packed WDM grid. A single window shape is shifted across frequency to define the channel index m , and each shifted window is then modulated in time to define the atom indexed by n . This is why one WDM coefficient can be read as the strength of one localized time-frequency pattern, rather than as the amplitude of a globally dsupported sinusoid.

More concretely, a single coefficient w_{nm} answers the question: “how much does the data resemble the atom centered at time bin n and frequency bin m ?” A large magnitude means that the data contains a feature with roughly the same time localization, central frequency, and oscillatory phase structure as that atom. Reading the WDM plane coefficient-by-coefficient in this way is often much more intuitive than interpreting a global Fourier series, because the index pair (n, m) already carries an approximate time-frequency location.

In the next subsections, we will provide formulae for the w_{nm} coefficients for both the interior, DC and

nyquist channels separately. In our final subsection, we investigate the orthogonal nature of the Wilson-basis.

1. Interior channels ($0 < m < N_f$)

Starting from Eq. (10) with the interior-channel basis element from Eq. (11), the forward transform can be evaluated in the Fourier domain using

$$w_{nm} = \sum_{l=-N/2}^{N/2-1} \tilde{x}_l \tilde{g}_{nm}^*(f[l]), \quad (16)$$

where $\tilde{x}_l = \tilde{x}[l]$ is the DFT of the input data with zeroth bin of $\tilde{x}$ placed in the middle of the spectrum. For a real-valued input signal, $\tilde{x}_{-l} = \tilde{x}_l^*$, and the real-valued window satisfies $\tilde{\varphi}[-l] = \tilde{\varphi}[l]$. Splitting the sum into positive and negative frequency halves and using these symmetries, the negative-frequency contribution is the complex conjugate of the positive-frequency contribution (see Appendix B for details), giving

$$\begin{aligned} w_{nm} &= \sqrt{2} (-1)^{nm} \\ &\times \Re \left[C_{nm}^* \sum_{l'=-N_t/2}^{N_t/2} \tilde{x}_{l'+mN_t/2} e^{2\pi i l' n / N_t} \tilde{\varphi}[l'] \right], \end{aligned} \quad (17)$$

where $l' = l - mN_t/2$ and $\Delta f = 1/T$.

The frequency shift $l = l' + mN_t/2$ contributes a phase $e^{2\pi i (mN_t/2) n / N_t} = e^{i\pi mn} = (-1)^{mn}$, so the result can be written compactly as

$$w_{nm} = \sqrt{2} (-1)^{nm} \Re C_{nm}^* x_m[n], \quad (18)$$

in this paper the $d = 1$ window provides adequate frequency separation while keeping the implementation simple and the time-domain filter short.

² Our definitions of the Wilson basis functions (11) carefully include the $m = 0$ and $m = N_f$ in order for orthogonality to hold.

where

$$x_m[n] = \sum_{l=-N_t/2}^{N_t/2-1} e^{2\pi i l n / N_t} X[l + mN_t/2] \tilde{\varphi}[l] \quad (19)$$

is the windowed, shifted inverse DFT of length N_t . The quantity $x_m[n]$ can be computed using a length- N_t FFT for each channel m , so the total cost of the interior channels is $N_f \times \mathcal{O}(N_t \log N_t)$.

2. DC channel ($m = 0$)

The DC edge channel has a doubled time-shift exponent and no C_{nm} factor in the basis definition (11). Following the same Fourier-domain evaluation yields

$$w_{n0} = \sqrt{2} \Re \left[\sum_{l=1}^{N_t/2} \tilde{x}_l e^{4\pi i l n / N_t} \tilde{\varphi}[l] \right] + \frac{1}{\sqrt{2}} \tilde{x}_0 \tilde{\varphi}[0]. \quad (20)$$

Note the factor of 4π in the exponent (versus 2π for interior channels): the DC basis element oscillates at twice the rate in the time index because it sees both positive and negative frequency copies of the window centered at $f = 0$.

3. Nyquist channel ($m = N_f$)

The Nyquist edge channel is handled analogously to the DC channel but centered at the Nyquist frequency $f_{\max}$. Its structure mirrors the DC case with appropriate frequency shifts. In practice, the Nyquist channel is computed using the same windowed-FFT machinery as the interior channels, with the frequency shift set to $mN_t/2 = N/2$.

$$w_{nN_f} = \sqrt{2} \Re \left[\sum_{l=N/2-N_t/2}^{N/2-1} \tilde{x}_l e^{4\pi i l n / N_t} \tilde{\varphi}[l - N_t/2] \right] + \frac{1}{\sqrt{2}} \tilde{x}_{N/2} \tilde{\varphi}[0]. \quad (21)$$

Collecting the results of each of the w_{nm} coefficients above, the forward transform reads

$$w_{nm} = \sqrt{2}(-1)^{mn} \Re \begin{cases} \sum_{l=1}^{N_t/2-1} \tilde{x}[l] \tilde{\varphi}[l] e^{4\pi i l n / N_t} + \frac{1}{2} \tilde{x}[0] \tilde{\varphi}[0], & m = 0, \\ C_{nm}^* \sum_{l'=-N_t/2}^{N_t/2-1} \tilde{x}[l' + mN_t/2] \tilde{\varphi}[l'] e^{2\pi i l' n / N_t}, & 1 \leq m \leq N_f - 1, \\ \sum_{l=N/2-N_t/2}^{N/2-1} \tilde{x}[l] \tilde{\varphi}[l - N/2] e^{4\pi i l n / N_t} + \frac{1}{2} \tilde{x}[N/2] \tilde{\varphi}[0], & m = N_f, \end{cases} \quad (22)$$

holding true for $N_f \in 2\mathbb{N}$.

C. Backward Transform

From the orthogonality condition of the Wilson basis, the inverse WDM transform is for the Fourier domain $\tilde{x}$ is immediate

$$\tilde{x}_l = \sum_{n=0}^{N_t-1} \sum_{m=0}^{N_f} \tilde{g}_{nm}[l] w_{nm}, \quad (23)$$

found via direct computation

$$\sum_{n=0}^{N_t-1} \sum_{m=0}^{N_f} \tilde{g}_{nm}[l] w_{nm} = \sum_{n=0}^{N_t-1} \sum_{m=0}^{N_f} \tilde{g}_{nm}[l] \sum_{l'=0}^{N-1} \tilde{x}[l'] \tilde{g}_{nm}^*[l'] \quad (24)$$

$$= \sum_{l'=0}^{N-1} \tilde{x}[l'] \delta_{ll'} = \tilde{x}[l]$$

where the first equality uses Eq. (10) and the second using Eq. (15).

Specifically, the sum over n produces $N_t \delta_{ll'}$ factors via the discrete orthogonality of the complex exponentials, and the sum over m covers the full frequency axis because adjacent windows satisfy $\tilde{\varphi}^2(f) + \tilde{\varphi}^2(f - \Delta F) =$

$1/\Delta F$ in their overlap region (proven in Appendix A 2). Together these two identities ensure exact round-trip reconstruction: forward followed by inverse returns the

original data to machine precision, as verified numerically in Section IV. We derive explicit formulae for the backward transformations given below B 4

$$\tilde{x}_m[l] = \frac{1}{\sqrt{2}} \begin{cases} \tilde{\varphi}[l] \sum_{n=0}^{N_t-1} w_{n0} e^{-4\pi i n l / N_t}, & m = 0, \quad 0 \leq l < N_t/2, \\ \tilde{\varphi}[l - mN_t/2] \sum_{n=0}^{N_t-1} C_{nm} w_{nm} e^{-2\pi i n l / N_t}, & 1 \leq m \leq N_f - 1, \quad (m-1)\frac{N_t}{2} \leq l < (m+1)\frac{N_t}{2}, \quad \bullet \\ \tilde{\varphi}[l - N/2] \sum_{n=0}^{N_t-1} w_{nN_f} e^{-4\pi i n l / N_t}, & m = N_f, \quad N/2 - N_t/2 \leq l \leq N/2, \end{cases} \quad (25)$$

where the original Fourier domain expression $\tilde{x}[l]$ on the output data is given via

$$\tilde{x}[l] = \sum_{m=0}^{N_f} \tilde{x}^m[l] \quad (26)$$

Equation (25) is the first time that the inverse WDM transform has been given in the literature to date.

III. STATISTICS WITHIN THE WDM FORMALISM

A. The WDM log-likelihood

For a Gaussian time-domain noise process characterized by a covariance matrix $\Sigma = \mathbb{E}[\mathbf{x}\mathbf{x}^T]$, the log-likelihood of the data $\mathbf{d}$ given a signal model $\mathbf{h}$, characterised via residuals $\mathbf{x} = \mathbf{d} - \mathbf{h}$ is

$$\ln p(\mathbf{d}|\mathbf{h}) = -\frac{1}{2} \mathbf{x}^T \Sigma^{-1} \mathbf{x} - \frac{1}{2} \ln \det \Sigma - \frac{N}{2} \ln 2\pi. \quad (27)$$

Following [4] with DFT convention such that

$$P_{kp} = \frac{1}{\sqrt{N}} e^{-\frac{2\pi i k p}{N}}, \quad \text{Unitary DFT matrix} \quad (28a)$$

$$\tilde{\mathbf{x}} = \sqrt{N} P \tilde{\mathbf{x}}, \quad \text{Forward FT} \quad (28b)$$

$$\mathbf{x} = \frac{1}{\sqrt{N}} \mathbf{P}^\dagger \tilde{\mathbf{x}}, \quad \text{Backward FT} \quad (28c)$$

$$\Sigma = \frac{1}{N} \mathbf{P}^\dagger \tilde{\Sigma} \mathbf{P}, \quad \text{TD Covariance} \quad (28d)$$

where the last equation is proved in [4] and where $\mathbf{P}$ is a unitary matrix and $\tilde{\Sigma} = \mathbb{E}[\tilde{\mathbf{x}}\tilde{\mathbf{x}}^\dagger]$ is the FD noise covariance matrix. We can substitute the expressions directly above into (27) to find the frequency domain

likelihood

$$\ln \mathcal{L}(\mathbf{d}|\boldsymbol{\theta}) = -\frac{1}{2} \tilde{\mathbf{x}}^\dagger \tilde{\Sigma}^{-1} \tilde{\mathbf{x}} - \ln \det \left(\frac{1}{N} \tilde{\Sigma} \right) - \frac{N}{2} \ln(2\pi). \quad (29)$$

By asserting that the time-domain noise covariance matrix is stationary and circulant, the frequency domain noise covariance matrix returns a purely diagonal matrix. In this section, we will perform a similar calculation to [4] and transform the frequency domain likelihood (29) into the WDM likelihood used for parameter inference.

The WDM coefficients w_{nm} are pixels that live on a grid of dimension $(N_t, N_f + 1)$. Here the grid is over-determined as we are also accounting for the zeroth and Nyquist bins, hence $N_f + 1$. Now let us define a bijection (isomorphism for nerds)

$$\alpha : \{n\}_{n=0}^{N_t-1} \times \{m\}_{m=0}^{N_f} \longrightarrow \{0, 1, \dots, N_t(N_f + 1) - 1\} \quad (30)$$

$$\alpha(n, m) \longmapsto \alpha = \{0, 1, \dots, n(N_f + 1) + m\}, \quad (31)$$

where, for any choice of $\alpha = n(N_f + 1) + m$ we have the real vector $\mathbf{w} \in \mathbb{R}^{N_t(N_f+1)}$. In other words, take the WDM coefficients (matrix) and flatten the matrix row by row to create a real vector.

We can then define the complex rectangular matrices $\mathbf{G}, \mathbf{G}^\dagger \in \mathbb{C}^{N_t(N_f+1) \times N}$ with linear maps

$$\mathbf{G} : \mathbb{C}^N \longrightarrow \mathbb{C}^{N_t(N_f+1)} \quad (32)$$

$$\mathbf{x} \longmapsto \mathbf{G}\tilde{\mathbf{x}} = \mathbf{w}, \quad (33)$$

$$(34)$$

$$\mathbf{G}^\dagger : \mathbb{C}^{N_t(N_f+1)} \longrightarrow \mathbb{C}^N$$

$$\tilde{\mathbf{w}} \longmapsto \mathbf{G}^\dagger \mathbf{w} = \tilde{\mathbf{x}}, \quad (35)$$

i.e., with action on the frequency domain vector $\tilde{\mathbf{x}} \in \mathbb{C}^N$ gives $\mathbf{w} = \mathbf{G}\tilde{\mathbf{x}}$ with inverse transform $\tilde{\mathbf{x}} = \mathbf{G}^\dagger \mathbf{w}$. In

component form, one should interpret $(\mathbf{G})_{\alpha,l} = \tilde{g}_{nm}^*[l]$, with components of $(\mathbf{G})_{\alpha,l}$ here read as row $\alpha \mapsto (n, m)$ evaluated at frequency index l . In component form:

$$(\mathbf{G}^\dagger \mathbf{G})_{ll'} = \sum_{n=0}^{N_t-1} \sum_{m=0}^{N_f} \tilde{g}_{nm}^*[l] \tilde{g}_{nm}[l'] = \delta_{ll'}, \quad (36)$$

$$(\mathbf{G}\mathbf{G}^\dagger)_{(n,m),(p,q)} = \sum_{l=-N/2}^{N/2-1} \tilde{g}_{nm}^*[l] \tilde{g}_{pq}[l] \quad (37)$$

$$= \delta_{m,q} \begin{cases} \frac{1}{2} (\delta_{n,p} + \delta_{n,p \pm N_t/2}), \\ \delta_{n,p}, \end{cases} \quad (38)$$

[Giorgio: That's a duplicate of (15), in the matrix formalism, so maybe we can move this formula to appendix] for the first case in Eq. (38) the zeroth $m = 0$ and $m = N_f$ Nyquist bins and second case the interior bins $m \in \{1, \dots, N_f - 1\}$. From the orthogonal conditions Eqs. (36, 38), one can see that $\mathbf{G}^\dagger \mathbf{G} = \mathbb{I}_N \neq \mathbf{G}\mathbf{G}^\dagger$.

Although $\mathbf{G}$ satisfies $\mathbf{G}^\dagger \mathbf{G} = \mathbb{I}_N$ and therefore admits an exact left-inverse, the opposite product $\mathbf{G}\mathbf{G}^\dagger$ is not the identity: by the first case of Eq. (38) it is an orthogonal projector onto the N -dimensional image of $\mathbf{G}$, with orthogonal complement spanned by the N_t redundant directions of the packed layout. The forward map $\mathbf{w} = \mathbf{G}\tilde{\mathbf{x}}$ embeds the signal space $\mathbb{C}^N$ into the larger packed space $\mathbb{C}^{N_t(N_f+1)}$, and the left-inverse $\mathbf{G}^\dagger$ recovers the signal,

$$\tilde{\mathbf{x}} \xrightarrow{\mathbf{G}} \mathbf{w} = \mathbf{G}\tilde{\mathbf{x}} \xrightarrow{\mathbf{G}^\dagger} \mathbf{G}^\dagger \mathbf{G}\tilde{\mathbf{x}} = \tilde{\mathbf{x}}, \quad (39)$$

so the forward-then-inverse round-trip is exact for every signal. The reverse round-trip, by contrast, fails for an arbitrary packed array $\mathbf{w}$ that is not itself the transform of a signal:

$$\mathbf{w} \xrightarrow{\mathbf{G}^\dagger} \mathbf{G}^\dagger \mathbf{w} \xrightarrow{\mathbf{G}} \mathbf{G}\mathbf{G}^\dagger \mathbf{w} \neq \mathbf{w}, \quad (40)$$

[Giorgio: Maybe here I would put an underline to the last inequality. It's almost always equal, but the 0 and Nf bins] since $\mathbf{G}\mathbf{G}^\dagger$ projects onto $\text{im}(\mathbf{G})$ and discards any component lying in the N_t -dimensional cokernel. Such spurious $\mathbf{w}$ do not arise in practice, where packed arrays are always obtained by transforming a genuine frequency-domain signal.

Separately, and on the practical side, the exactness of the forward-then-inverse round-trip relies on $\mathbf{G}^\dagger \mathbf{G} = \mathbb{I}_N$, which in turn requires the DC and Nyquist edge channels to be included and implemented correctly. Omitting or mishandling these channels breaks the completeness relation and introduces non-trivial reconstruction errors in $\tilde{\mathbf{x}} \mapsto \mathbf{w} \mapsto \tilde{\mathbf{x}}$.

Consider the two

$$\tilde{\Sigma} = \mathbb{E}[\tilde{\mathbf{x}}\tilde{\mathbf{x}}^\dagger] = \mathbf{G}^\dagger \mathbb{E}[\mathbf{w}\mathbf{w}^\dagger] \mathbf{G} = \mathbf{G}^\dagger \Sigma_w \mathbf{G} \quad (41)$$

$$\Sigma_w = \mathbb{E}[\mathbf{w}\mathbf{w}^\dagger] = \mathbf{G} \mathbb{E}[\tilde{\mathbf{x}}\tilde{\mathbf{x}}^\dagger] \mathbf{G}^\dagger = \mathbf{G} \tilde{\Sigma} \mathbf{G}^\dagger \quad (42)$$

for $\Sigma_w \in \mathbb{R}^{N_t(N_f+1) \times N_t(N_f+1)}$ the real valued WDM noise covariance matrix.

Now, for stationary and circulant noise we have that $\tilde{\Sigma}$ is trivially invertible. This does not imply that the product $\mathbf{G}\Sigma_w\mathbf{G}^\dagger$ is invertible. One can show this using the following identities

$$\text{Rank}(\mathbf{A}^\dagger) = \text{Rank}(\mathbf{A}) \quad (43a)$$

$$\text{Rank}(\mathbf{A}^\dagger \mathbf{A}) = \text{Rank}(\mathbf{A}) \quad (43b)$$

$$\text{Rank}(\mathbf{A}\mathbf{B}\mathbf{C}) \leq \min(\text{Rank}(\mathbf{A}), \text{Rank}(\mathbf{B}), \text{Rank}(\mathbf{C})) \quad (43c)$$

From (43a) and (43b), we see that $N = \text{Rank}(\mathbf{G}^\dagger \mathbf{G}) = \text{Rank}(\mathbf{G}) = \text{Rank}(\mathbf{G}^\dagger)$. Similarly, we have from Eq.(43c) that $\text{Rank}(\Sigma_w) \leq N$. Since the matrix Σ_w does not have full rank, i.e., $\text{Rank}(\Sigma_w) \neq (N + N_t)$, we deduce that Σ_w is not invertible over the full span of pixels $n \in \{0, \dots, N_t - 1\}$ and $m \in \{0, \dots, N_f\}$.

To make progress, we define the Moore-Penrose pseudo inverse Σ_w^+ of the matrix Σ_w

$$\Sigma_w^+ = \mathbf{G} \tilde{\Sigma}^{-1} \mathbf{G}^\dagger. \quad (44)$$

Where one can immediately check the following identities **[Giorgio: How about just stating that $\Sigma_w^+ \Sigma_w = \Sigma_w \Sigma_w^+ = 1$ here below? Also, we may write it in-line and not with an equation, it's quite easy to see it, right?][Ollie: They only equal 1 over the interior bins. Including the zeroth and Nyquist, the computation returns $\mathbf{G}\mathbf{G}^\dagger$.]**

$$\Sigma_w^+ \Sigma_w \Sigma_w^+ = \Sigma_w^+, \quad (45a)$$

$$\Sigma_w \Sigma_w^+ \Sigma_w = \Sigma_w, \quad (45b)$$

$$(\Sigma_w \Sigma_w^+)^\dagger = \Sigma_w \Sigma_w^+, \quad (45c)$$

$$(\Sigma_w^+ \Sigma_w)^\dagger = \Sigma_w^+ \Sigma_w. \quad (45d)$$

since $\mathbf{G}^\dagger \mathbf{G} = \mathbb{I}_N$. To build the WDM likelihood, we substitute $\tilde{\mathbf{x}} = \mathbf{G}^\dagger \mathbf{w}$ into (29) to obtain

$$\ln \mathcal{L}(d|\boldsymbol{\theta}) = -\frac{1}{2} \mathbf{w}^T \Sigma_w^+ \mathbf{w} \quad (46)$$

$$- \ln \det \left(\frac{1}{N_f N_t} \mathbf{G}^\dagger \Sigma_w \mathbf{G} \right) - \frac{N_f N_t}{2} \ln(2\pi). \quad (47)$$

With pseudo-inverse of the WDM matrix given by (44). The singular nature of the WDM covariance matrix arises due to the overcomplete nature of the

packed basis when including $m = 0$ and $m = N_t$. In other words, the zeroth and Nyquist bins are dependent on one another, forcing linear dependence between the rows. A proper treatment of the likelihood would require both a pseudo-inverse of the WDM matrix and a pseudo-determinant using the SVD algorithm – both rather costly for the sake of two extra bins. However, it is far more computationally convenient to work with the interior atoms $1 \leq m \leq N_f$ which results in square matrices $\mathbf{G}^{\text{int}} \in \mathbb{C}^{N \times N}$ and $\mathbf{\Sigma}_w^{\text{int}} \in \mathbb{R}^{N \times N}$ with full rank – i.e., now a zero dimensional co-kernel. Similarly, over the interior bins we have that $\mathbf{G}\mathbf{G}^\dagger = \mathbf{G}^\dagger\mathbf{G} = \mathcal{I}_N$ and $(\mathbf{\Sigma}_w^{\text{int}})^+ = (\mathbf{\Sigma}_w^{\text{int}})^{-1}$. Hence dropping the DC and Nyquist bins yields allows for a simplification of the log-likelihood

$$\begin{aligned} \ln \mathcal{L}(d|\boldsymbol{\theta}) = & -\frac{1}{2} \mathbf{w}^T (\mathbf{\Sigma}_w^{\text{int}})^{-1} \mathbf{w} \\ & - \ln \det \left(\frac{1}{N_f N_t} \mathbf{\Sigma}_w^{\text{int}} \right) - \frac{N_f N_t}{2} \ln(2\pi). \end{aligned} \quad (48)$$

It is interesting to note that the quadratic or log-determinant piece can be evaluated in any domain – time domain, frequency domain or WDM domain translated via matrix-algebra. For example, if fast signal templates in the WDM domain were to hand and one were interested in noise estimation in a purely stationary context, one could compute the quadratic piece in the WDM domain and noise term in the frequency domain.

B. Local stationary noise

At this point we place the condition on the time-domain noise covariance matrix that it is both stationary and circulant. Such conditions give a purely diagonal noise covariance matrix with entries proportional to the covariance matrix

$$\tilde{\mathbf{\Sigma}}_{ll'} = \frac{NS[l]}{2\Delta t} \delta_{ll'} \quad (50)$$

for S the noise Power Spectral Density, a function relating the power of instrumental (and/or galactic) noise with frequency. White noise is governed via $S[l] \propto \text{const}$ and coloured noise otherwise. and, since

$$\mathbf{\Sigma}_w^{\text{int}} = (\mathbf{G}^{\text{int}})^\dagger \tilde{\mathbf{\Sigma}} \mathbf{G}^{\text{int}} \quad (51)$$

We therefore have in component form

$$(\mathbf{\Sigma}_w^{\text{int}})_{\alpha\beta} = \frac{N}{2\Delta t} \sum_{l,l'} \tilde{g}_{nm}^*[l] \tilde{g}_{pq}[l] S[l] \quad (52)$$

At this point, we will assume local stationarity such that $S_n[l] \approx S_n[k, l]$ for time and frequency index (k, l) .

In other words, S_n is now an evolutionary power spectral density which could evolve slowly over time. Provided the evolutionary power spectral density evolves slowly over a wavelet bandwidth ΔF , we can pull the evolutionary PSD out of the summation [Giorgio: Here is, for me, the most important point over all the stats things. What is the formal way to relate $S_n[k, l]$ and $S_n[l]$? is it $S_n[k, l] = S_n[l]$? If yes, sorry to be pedantic, but I feel like we have to prove it. Also, we should change notation, in order not to confuse the reader][Ollie: Yeah, I think we change notation – at least get rid of the n on the subscript (sorry, my bad before – you were doing the right thing ignoring n . I think the evolutionary PSD comes from Taylor expanding the PSD itself as a function of time $t = t_0$ and, provided $\dot{S}[n, m]|_{n=n_0}$ is small compared to the time evolution over the bin ΔF , it can be dropped. Something like this? I can try and prove it?] [Giorgio: Yes, something like it. I would be happy to see something that takes the last line of the equation below, and relate S_{kl}^{WDM} to $S_n[l]$, like $S_{kl}^{\text{WDM}} = S_n[l] \times \text{something}$. I believe there is no need to introduce $S_n[k, l]$, we can choose one between $S_n[k, l]$ and S_{kl}^{WDM} . This will save also another (not really useful, I believe) equation that relates $S_n[k, l]$ and S_{kl}^{WDM} .]

$$\begin{aligned} (\mathbf{\Sigma}_w^{\text{int}})_{\alpha\beta} & \approx \frac{N}{2\Delta t} S_n[k, l] \sum_{l, l'} \tilde{g}_{nm}^*[l] \tilde{g}_{pq}[l'] \\ & = \frac{N}{2\Delta t} S_n[k, l] \delta_{np} \delta_{mq} \end{aligned} \quad (53)$$

$$= \delta_{nm} \delta_{pq} S_{kl}^{\text{WDM}} \quad (54)$$

where we have restricted the sum over the interior bins $m \in \{1, \dots, N_f - 1\}$. Here S_{nm}^{WDM} is defined by

$$S_{nm}^{\text{WDM}} = \frac{N}{2\Delta t} S_n[n, m] \quad (55)$$

$$= (NN_f) S_n[n, m] \Delta F \quad (56)$$

the evolutionary PSD S_{nm}^{WDM} – the noise PSD with the freedom to evolve as a function of both frequency f_m and time t_n . We remark that the evolutionary PSD is not a covariance matrix, but only when combined with the Kronecker deltas above as a result of local-stationarity can it be used as a covariance matrix. The factor NN_f relative to the natural pixel variance $S_n \Delta F$ arises from our discrete-DFT convention (Eq. 3), in which $\tilde{x}[l]$ does not carry an explicit factor of Δt . Cornish [5] uses Δt -normalised transforms and accordingly obtains $S_{nm}^{\text{WDM}} \approx S_n(f_m) \Delta F$.

If we include the zeroth and Nyquist frequency bins $m = \{0, N_f\}$ and using the first equation in (15), we see that the WDM covariance at these bins takes a slightly

different form

$$(\Sigma_w)_{\alpha\beta}^{m \in \{0, N_f\}} = \frac{1}{2} S_{nm}^{\text{WDM}} \delta_{mq} (\delta_{np} + \delta_{n, p \pm N_t/2}). \quad (57)$$

implying that the variance of the zeroth and Nyquist bins is different from the interior bins. This is analogous to the frequency domain analogue as described in Appendix A of [4].

Over the interior bins only, substituting the diagonal covariance (54) into the general Gaussian likelihood yields the WDM Whittle-based log-likelihood:

$$\ln p(d|h) \propto - \sum_{\substack{n \\ m \in \text{int}}} \left[\ln \left(\frac{1}{N_t N_f} S_{nm}^{\text{WDM}} \right) + \frac{1}{2} \frac{(d_{nm} - h_{nm})^2}{S_{nm}^{\text{WDM}}} \right], \quad (58)$$

[Giorgio: Is there a - sign missing in front of $\frac{1}{2}$ in this equation above? Also, why did you split N to $N_t N_f$?] where d_{nm} , h_{nm} are the WDM coefficients of the data and signal model, respectively. The likelihood (58) is sufficient for both noise and signal parameter estimation. From this point onwards, we assume a known noise covariance matrix in the WDM domain, implying that the likelihood (58) is proportional up to the quadratic piece (second term).

In the ideal orthogonal limit, whitened noise – noise with $S_n[l] \propto \text{const.}$ across the support of each atom – reduces this covariance to diagonal form. In that case the WDM coefficients are uncorrelated, and for Gaussian noise they are therefore independent. In the sampled packed implementation used here, it is useful to distinguish two statements. First, the interior atoms ($m = 1, \dots, N_f - 1$) satisfy the discrete orthonormality relation to machine precision. Second, the packed real-valued storage includes special DC and Nyquist edge channels, so the full stored coefficient array is best viewed as a convenient packed representation rather than a minimal orthonormal basis. For this reason, the decorrelation discussion below focuses on the interior coefficients. For whitened noise their empirical correlations remain very small, while for coloured, non-stationary, or non-Gaussian noise off-diagonal structure becomes more visible and the coefficients should be regarded as only approximately decorrelated rather than strictly independent. Figure 2 illustrates these points numerically for a small sampled transform and for white and coloured noise realizations. The practical pay-off of the (approximate) orthonormality and decorrelation discussed above is that standard gravitational-wave data analysis quantities — the likelihood, signal-to-noise ratio, and Fisher information matrix — can all be written as simple sums over WDM pixels, in direct analogy with the familiar frequency-domain expressions.

C. The WDM Inner Product

In a similar way to a frequency domain analysis, one can define the noise-weighted inner product in the WDM domain

$$(a|b)_{\text{WDM}} = \mathbf{a}^T (\Sigma_w^{\text{int}})^{-1} \mathbf{b} \quad (60)$$

$$= \sum_{\substack{n \\ m \in \text{int}}} \frac{a_{nm} b_{nm}}{S_{nm}}, \quad (61)$$

which reduces to the frequency domain inner product via matrix algebra

$$(a|b)_{\text{WDM}} = \mathbf{a}^T (\Sigma_w^{\text{int}})^{-1} \mathbf{b} \quad (62)$$

$$= \tilde{\mathbf{a}}^\dagger \mathbf{G}^\dagger \mathbf{G} (\tilde{\Sigma}_{\text{int}})^{-1} \mathbf{G} \mathbf{G}^\dagger \mathbf{b} \quad (63)$$

$$= \tilde{\mathbf{a}}^\dagger (\tilde{\Sigma}_{\text{int}})^{-1} \tilde{\mathbf{b}} \quad (64)$$

$$= \frac{4N}{\Delta t} \text{Re} \left(\sum_{m=1}^{N_f-1} \frac{\tilde{a}^*[m] \tilde{b}[m]}{S_n[m]} \right) \quad (65)$$

$$= (a|b)_{\text{freq}}. \quad (66)$$

Where each vector and Wilson basis matrix $\mathbf{G}$ only contains full time information $n \in \{0, \dots, N_t - 1\}$ but excluding the zeroth/Nyquist bin $m \in \{1, \dots, N_f - 1\}$. giving

$$\ln p(d|h) \propto -\frac{1}{2} (d - h | d - h)_{\text{WDM}}. \quad (67)$$

Resulting in This expression is structurally identical to the standard frequency-domain Whittle likelihood, but with the one-dimensional sum over frequency bins replaced by a two-dimensional sum over time-frequency pixels. The key advantage is that S_{nm} can vary with time, naturally accommodating non-stationary noise without requiring off-diagonal terms in the covariance.

The advantage of writing (67) in terms of the noise-weighted inner product is that familiar quantities used throughout GW astronomy immediately follow. For example, the SNR,

$$\rho^{\text{opt}} = (h|h)_{\text{WDM}}, \quad \text{SNR} \quad (68)$$

$$\rho(h) = \frac{(d|h)_{\text{WDM}}}{\sqrt{(h|h)_{\text{WDM}}}}, \quad \text{Effective SNR} \quad (69)$$

$$\Gamma_{ij} = (\partial_i h | \partial_j h)_{\text{WDM}}, \quad \text{Fisher Matrix} \quad (70)$$

$$\Delta \theta^i = (\Gamma^{-1})^{ij} (\partial_i h | n + \delta h)_{\text{WDM}}. \quad \text{Cutler Valisneri} \quad (71)$$

[Giorgio: Shall we reference the paper of the last one?] In other words, any familiar frequency domain quantity that involves an inner product $(a|b)_{\text{freq}}$ can equivalently be written down in the WDM domain via $(a|b)_{\text{WDM}}$.

These expressions assume that the diagonal approximation (54) holds. When it does not — for example, when the noise PSD varies rapidly compared to ΔT , or when the noise is strongly non-Gaussian — the off-diagonal covariance terms must be retained and the simple pixel-wise sums above become inadequate. Figure 2 provides a practical diagnostic: if the off-diagonal entries of the empirical covariance remain small after whitening, the diagonal likelihood (58) is a safe approximation. A natural question comes to light — when does the diagonal approximation hold?

The WDM tiling obeys $\Delta T \Delta F = 1/2$ (see Eq. (7)), so each atom occupies the minimum allowed time-frequency area consistent with the uncertainty principle, though the actual time-bandwidth product of the cosine-tapered Meyer window exceeds this minimum. The diagonal covariance (54) holds whenever the noise PSD varies slowly compared to this joint support. Concretely, this is the condition of local stationarity: spectral variations on timescales shorter than ΔT , or spectral features narrower than ΔF , will induce off-diagonal covariance terms. The WDM tiling therefore poses an inherent trade-off: choosing finer frequency resolution (larger ΔT) suppresses spectral leakage within each atom but degrades temporal tracking of non-stationarity. In practice a single choice of N_f often suffices for mildly non-stationary noise, but data with sharp spectral features or rapid temporal variation may require a multi-resolution strategy, as discussed by Cornish [5].

IV. WDM_TRANSFORM PACKAGE

[Ollie: I also think a sexy flowchart describing the package would be amazing here. At least with all of the various modules. The main thing we are selling here is (1) an understanding of the mathematical formalism (done) and (2) the code base. A simple flow chart so that any person reading the paper in 15 seconds can stop on would be really useful. At least then they can see: forward/backwards transforms, docs, unittests and the important modules.] [Avi: sure can do..]

The `wdm_transform` package provides NumPy and JAX backends for the Wilson–Daubechies–Meyer (WDM) transform, both implementing the packed $(N_t, N_f + 1)$ coefficient layout and forward/inverse conventions described above. The primary interface exposes the full forward and inverse transforms; in addition, band-limited routines allow the transform to be evaluated over a specified subset of frequency channels, reducing compute cost when only a portion of the time-frequency plane is needed.

From the implementation point of view, the dominant operations in the forward transform are one length- N FFT of the input data together with one length- N_t sub-band transform for each stored channel. This gives a cost of

$$T_{\text{forward}} = \mathcal{O}(N \log N) + \mathcal{O}(N \log N_t), \quad (72)$$

which is asymptotically $\mathcal{O}(N \log N)$ because $N = N_t N_f$ and $N_t \leq N$. The inverse transform has the same scaling up to constant factors: it reconstructs the full spectrum from the packed channels and then applies one inverse FFT back to the time domain. In both directions, the stored coefficient grid contains $\mathcal{O}(N)$ real numbers, and the temporary working arrays used by the implementation are also linear in N , so the overall memory footprint is $\mathcal{O}(N)$.

We validate the implementation using round-trip reconstruction tests. For an input time series x , we compute its packed WDM coefficients, apply the inverse transform, and compare the reconstructed series $\hat{x}$ with the original. Across the benchmarked problem sizes and backends, the relative reconstruction error

$$\epsilon_{\text{rt}} = \frac{\|\hat{x} - x\|_2}{\|x\|_2} \quad (73)$$

remains at the level of double-precision floating-point roundoff, $\epsilon_{\text{rt}} \lesssim 10^{-16}$, providing a direct numerical check that the packed layout, including the DC and Nyquist edge channels, is implemented consistently across both backends.

All timings are drawn from the benchmark suite included in the package documentation. Each data point reports the median wall-clock time over 7 repeated runs in 64-bit floating-point arithmetic; JAX JIT compilation is excluded via a single warmup evaluation, and GPU timings assume arrays already resident on the device (host–device transfer is not included). Benchmarks were performed on Google Colab using a CPU node (2× Intel Xeon vCPU, 2.0–2.2 GHz, 12.7 GB RAM) and a GPU node (NVIDIA T4, 16 GB VRAM).

Figure 3 shows the forward-kernel runtime as a function of input size N for the NumPy (CPU), JAX (CPU), and JAX (GPU) backends, together with the batched-execution speedup $t_{\text{serial}}/t_{\text{batch}}$. Here, t_{serial} is the wall-clock time for three independent single-channel calls and t_{batch} is the time for a single call that processes all three channels simultaneously.

At large N all three backends approach $\mathcal{O}(N \log N)$ scaling, consistent with the dominant length- N FFT in the forward transform. Deviations at small N reflect different overhead regimes: per-call Python and memory-allocation costs for NumPy, and kernel-dispatch latency for JAX. The JAX GPU runtime is nearly flat below $N \approx 10^5$, indicating that transforms at those sizes are

dominated by kernel-launch overhead rather than arithmetic throughput; beyond this threshold GPU efficiency improves and, at $N = 10^6$, the GPU scalar runtime (≈ 13 ms) is roughly $35\times$ lower than NumPy (≈ 460 ms).

The speedup panel shows that batched execution yields no measurable benefit for NumPy ($t_{\text{serial}}/t_{\text{batch}} \approx 1$ throughout), as NumPy does not exploit the extra batch dimension for additional parallelism. JAX (CPU) recovers $1.5\text{--}3\times$ via JIT-compiled vectorisation over the batch axis. JAX (GPU) achieves up to $\sim 4\times$ at small N , where all three channels fit simultaneously into the device pipeline; the speedup decreases to $\approx 1.5\times$ at $N = 10^6$ as individual transforms approach device saturation. The $\epsilon_{\text{rt}} \lesssim 10^{-16}$ round-trip errors established above hold for all backends and sizes; the runtime differences therefore reflect implementation efficiency rather than any loss of numerical fidelity.

These benchmarks characterise the present NumPy and JAX implementations under the conventions used in this paper, rather than providing an exhaustive comparison with existing pipeline-specific WDM codes. The main conclusion is that the JAX backend can substantially reduce the cost of large WDM transforms while preserving the same floating-point reconstruction accuracy as the NumPy implementation.

A. Resolved galactic-binary inference in a galactic background

[Giorgio: After discussing with Ollie, we think it is better to do the inference with just instrumental noise, which is stationary. Avi, what do you think about it? If you agree, could you please re-run the analysis? We can take care of the rewriting of the section (cutting off the galactic background formulas)] [Lore: consider looking into [16] for GBs in non stationary noise ;)] As an end-to-end inference example, we generated a one-year LISA-like dataset containing stationary instrumental noise, a stochastic galactic foreground, and one resolved galactic binary (GB). Both the frequency-domain and WDM-domain likelihoods use the same stationary noise model throughout; this is an intentional controlled comparison designed to isolate the effect of the WDM basis change from any modelling differences. The effective PSD $S_n^{\mathcal{I}}(f)$ in each channel $\mathcal{I}$ includes both the instrumental noise contribution and the time-averaged galactic foreground. Figure 4 shows the characteristic-strain comparison between the injected source and the channel noise model, and Figure 5 shows the overlaid posterior constraints from the two analyses.

The data set is constructed on a frequency grid $\in (10^{-4}, 3 \times 10^{-3} \text{ Hz})$. The unresolved galactic foreground

is described by the phenomenological PSD [17]

$$S_{\text{gal}}(f) = A_{\text{gal}} f^{-7/3} \exp \left[- \left(\frac{f}{f_{1,\text{gal}}} \right)^{\alpha_{\text{gal}}} \right] \times \left[1 + \tanh \left(\frac{f_{\text{knee,gal}} - f}{f_{2,\text{gal}}} \right) \right], \quad (74)$$

using a standard smooth fit for the LISA galactic foreground. To project this foreground into the detector channels we combine that frequency model with an anisotropic Milky Way sky map obtained by line-of-sight integration through the bulge-plus-disk density model [19]

$$\rho_{\text{gal}}(x, y, z) = \rho_0 \left[A e^{-r^2/R_b^2} + (1 - A) e^{-u^2/R_d^2} \text{sech}^2(z/Z_d) \right], \quad (75)$$

where $r^2 = x^2 + y^2 + z^2$ and $u^2 = x^2 + y^2$. We evaluate the time-dependent detector response using equal-arm LISA orbits [2] sampled at 90 orbital times, and use that response to map the anisotropic foreground into the A,E,T TDI channels, assuming that the spectrum is stationary within each of these 90 orbital segments. We then draw complex Gaussian realisations of both the instrumental noise and the chunked galactic foreground and stitch them into a full-year spectrum. The resolved source is injected afterward as a TDI 1.5 A,E,T waveform computed with `jaxgb` [3], using equal-arm LISA orbits and local Fourier support $n = 256$. In the realization shown here the cadence is $\Delta t = 166.7$ s, $T_{\text{obs}} = 1$ yr, and the injected binary has $f_0 = 1.3862897$ mHz, $\dot{f} = 1.32 \times 10^{-18} \text{ Hz s}^{-1}$, $A = 1.19 \times 10^{-23}$, and $\phi_0 = 1.442$ rad.

[Giorgio: I will define sigma gal, and check everything here, after Ollie is ready with the math of the statistics][Ollie: I think we are ready for you to proceed here.]

The resolved GB model uses the standard eight source parameters $(f_0, \dot{f}, A, \alpha, \delta, \psi, \iota, \phi_0)$, but for this controlled comparison we sample only $(f_0, \dot{f}, A, \phi_0)$ and hold sky position and orientation fixed to their injected values. The carrier frequency is drawn near the fixed reference value $f_{0,\text{ref}} = 1.38591$ mHz through $\delta \ln f_0 \sim \mathcal{U}[-10^{-3}, 10^{-3}]$, while $\dot{f}$ and A are drawn from truncated Gaussians in log space over $[5 \times 10^{-19}, 4 \times 10^{-18}] \text{ Hz s}^{-1}$ and $[6 \times 10^{-24}, 1.7 \times 10^{-23}]$, respectively.

The frequency-domain analysis uses a local A/E/T band centered on the prior support of the source and evaluates the Whittle likelihood

$$\ln \mathcal{L}_{\text{freq}}(\boldsymbol{\theta}) \propto - \sum_{\mathcal{I} \in \{A,E,T\}} \sum_k \left[\ln S_n^{(\mathcal{I})}[k] + \frac{2N |\tilde{d}^{(\mathcal{I})}[k] - \tilde{h}^{(\mathcal{I})}[k; \boldsymbol{\theta}]|^2}{\Delta t S_n^{(\mathcal{I})}[k]} \right], \quad (76)$$

where for each channel $\mathcal{I}$ and frequency bin k , $\tilde{d}^{(\mathcal{I})}[k]$ and $\tilde{h}^{(\mathcal{I})}[k; \theta]$ are the complex local Fourier modes and $S_n^{(\mathcal{I})}[k]$ is the stationary effective PSD, namely instrumental noise³ plus the time-averaged galactic foreground contribution in channel c . The WDM analysis starts from the same local Fourier patch, embeds the template into that patch, and applies the WDM transform with $N_t = 32$, $a = 1/3$, and $d = 1$. The resulting WDM likelihood is the diagonal Gaussian specialization of Eq. (58),

$$\ln \mathcal{L}_{\text{WDM}}(\theta) = -\frac{1}{2} \sum_{c \in \{A, E, T\}} \sum_{n, m} \left[\ln(2\pi\sigma_{c, m}^2) + \frac{\left(w_{c, nm}^{(d)} - w_{c, nm}^{(h)}(\theta)\right)^2}{\sigma_{c, m}^2} \right], \quad (77)$$

with a stationary per-pixel variance following from Eq. (??) with $\Delta F = f_{\text{Nyq}}/N_f$ over the interior $m \in \{1, \dots, N_f\}$

$$\sigma_{c, m}^2 = S_{nm}^{\text{WDM}} = \delta_{np} \delta_{pq} \frac{N S_c(f_m)}{2\Delta t}, \quad (78)$$

broadcast across the WDM time bins. In other words, this example uses the same stationary PSD in both domains and asks whether the WDM transform reproduces the frequency-domain posterior under that shared noise model. **[Avi: Leave a note here on]**

Both analyses use the NumPyro [23] implementation of the No-U-Turn Sampler with a dense mass matrix, target acceptance probability 0.85, and maximum tree depth 10. Each run uses 800 warmup steps followed by 1000 retained samples from a single chain, initialized near the injected parameters with small jitters in $\log \dot{f}$ and $\log A$. In the realization shown here, both samplers produced zero divergences. The latent effective sample sizes are 613–637 for the frequency-domain chain and 590–714 for the WDM chain, while the corresponding $\hat{R}$ values lie in the ranges 0.999–1.007 and 0.999–1.016. The truth-band optimal SNRs are 22.5469 in the frequency-domain calculation and 22.5462 in the WDM calculation. All reported 90% credible intervals contain the injected truth in both runs, and the posterior medians differ only by 9.0×10^{-11} Hz in f_0 , 2.2×10^{-20} Hz s⁻¹ in $\dot{f}$, 5.2×10^{-26} in A , 6.1×10^{-3} rad in ϕ_0 , and 0.10 in SNR. Within the numerical precision of this study, the WDM and frequency-domain analyses are therefore statistically indistinguishable for this seeded resolved-GB example.

Table I. Jensen–Shannon divergence between WDM and frequency-domain marginal posteriors across 100 independent noise realisations. The injected signal parameters are held fixed; only the noise draw varies. JSD is computed on the 1D marginal for each sampled parameter. All values in nats; the $[q_{05}, q_{95}]$ interval characterises seed-to-seed variability.

Parameter	Median	$[q_{05}, q_{95}]$
f_0 [Hz]	0.0139	[0.0086, 0.0469]
$\dot{f}$ [Hz/s]	0.0106	[0.0069, 0.0139]
A	0.0125	[0.0087, 0.0227]
ϕ_0 [rad]	0.0153	[0.0051, 0.0686]

To test whether the agreement seen in Figures 4 and 5 is robust across a range of signals and noise draws, we repeated the inference for 100 independent seeds. For each seed the source parameters ($f_0, \dot{f}, A, \phi_0$) and the sky position and orientation were drawn fresh from the prior, and an independent realisation of the instrumental noise and galactic foreground was generated; sky position and orientation were then held fixed at their injected values during sampling. Both likelihoods use the same time-averaged effective PSD throughout, so this study isolates the effect of the WDM basis change from any difference in noise modelling: the non-stationarity of the galactic foreground enters both analyses as the same shared systematic and does not contribute to the JSD. For each seed we computed the Jensen–Shannon divergence (JSD) between the WDM-domain and frequency-domain 1D marginal posteriors. The JSD is a symmetric, bounded measure of distributional similarity, with $\text{JSD} = 0$ for identical distributions. Table I summarises the resulting distributions across the 100 seeds: median values lie at or below 0.016 nat for all four parameters, and the 90th-percentile upper bound reaches at most 0.07 nat. These results confirm that the WDM-domain likelihood faithfully reproduces the frequency-domain posterior under a shared stationary noise model, across a representative population of signal parameters and noise realisations.

The `wdm_transform` package enables both a straightforward NumPy-based development path and high-performance accelerator execution via JAX. For small data lengths ($N \lesssim 2^{20}$), the NumPy version is competitive or even faster due to its extremely low overhead. However, once N exceeds $\sim 2^{20}$, the GPU-accelerated versions achieve markedly lower runtimes while both backends maintain round-trip reconstruction errors at floating-point roundoff. This combination of flexibility, accuracy, and performance makes `wdm_transform` a useful tool for WDM time-frequency analysis of long-duration gravitational-wave signals.

³ We adopt the Gaussian two-parameter model for the instrumental noise PSD [1], using Eqs. (B14) and (B15) therein.

V. CONCLUSIONS

We have presented a self-contained, pedagogical account of the Wilson–Daubechies–Meyer (WDM) wavelet-packet transform, building on the theoretical foundations established in prior work [5, 21? ?], and its implementation in the open-source `wdm_transform` package. The primary goal of this work is to make the discrete, sampled WDM implementation explicit and reproducible for practitioners new to the transform, by walking through the full three-case basis definition (DC, interior, and Nyquist channels), the packed (N_t, N_f+1) coefficient layout, the forward and inverse transforms with detailed derivations, and the conditions under which WDM coefficients may be treated as approximately independent.

On the implementation side, the NumPy and JAX backends both achieve exact round-trip reconstruction to floating-point precision. The JAX backend provides substantially lower runtimes for large transforms, making it practical to incorporate WDM-domain operations into inference pipelines that require millions of likelihood evaluations. **[Ollie: One point to make here – is the WDM transformation as cheap as a simple FFT? If so, then this makes incorporating the WDM methodology into any FD based analysis will not improve computational costs, implying that there is no costs to using WDM over current FD analyses? Does this make sense?] [Avi: not quite as cheap as a single FFT i think... the forward WDM is an FFT plus N_f windowed inverse FFTs of length N_t , so the scaling is $O(N \log N_t)$ rather than $O(N \log N)$. in practice the constant is small enough that the cost is comparable to a few FFTs... maybe we can include FFT timings in the runtime plot to compare.]**

Several limitations of the current work should be noted. First, the package implements only the $d = 1$ Meyer window; higher-order windows ($d > 1$) offer better spectral leakage suppression and may be preferable for data with sharp spectral features. Second, the paper does not address the generation of gravitational-wave signals directly in the WDM domain — where much of the computational advantage of the WDM representation lies [5] — and we defer fast WDM-domain waveform generation to future work. Third, the treatment of data gaps and edge effects, while conceptually straightforward, requires careful handling of the boundary basis elements and is left to downstream analyses built on top of the package.

Fourth, the LISA galactic-binary inference example of Section IV A uses a time-averaged effective PSD in both the frequency-domain and WDM-domain likelihoods; this controlled comparison isolates the WDM

basis change but does not exercise the principal advantage of the WDM representation, namely the ability to assign a time-varying S_{nm} to each pixel and thereby accommodate genuinely non-stationary noise without off-diagonal covariance terms. Demonstrating this advantage on realistic LISA non-stationary backgrounds is an important direction for future work.

Looking ahead, the most immediate extensions are the incorporation of fast WDM-domain waveform generation for compact binary signals, the implementation of evolutionary power spectral density estimation directly on the WDM grid, and integration of the package into existing LISA data analysis frameworks. The combination of a pedagogical reference, a tested open-source implementation, and GPU-accelerated performance should lower the barrier to adopting the WDM transform within modern, autodiff-compatible Python inference pipelines for the long-duration, non-stationary signals expected from future detectors such as LISA. Because the JAX backend is fully differentiable, gradient-based samplers (HMC, NUTS) and variational inference are available out of the box on top of the WDM likelihood, without any additional differentiation work by the user. Future work should focus on determining the optimal tiling of the WDM space for each source class. A systematic comparison with the short-time Fourier transform, assessing respective advantages and limitations, would help identify the most appropriate time–frequency representation and tiling for different types of sources.

DATA AND SOFTWARE AVAILABILITY

The software developed for this research is available on GitHub at https://github.com/pywavelet/wdm_transform. Documentation and worked examples are available in the repository documentation at https://github.com/pywavelet/wdm_transform/tree/main/docs. The online documentation includes minimal examples demonstrating the construction of a WDM tiling, application of the forward transform, and round-trip reconstruction with the inverse transform.

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Appendix A: Normality condition of the g_{nm} basis

To remind the reader, We introduce the basis $\tilde{g}_{nm}(f)$, given by eq.(11),

$$\tilde{g}_{nm}[l] = \frac{1}{\sqrt{2}} \begin{cases} e^{-4\pi i n l / N_t} \tilde{\varphi}[l], & m = 0, \\ e^{-2\pi i n l / N_t} \left(C_{nm} \tilde{\varphi}[l - mN_t/2] + C_{nm}^* \tilde{\varphi}[l + mN_t/2] \right), & 0 < m < N_f, \\ e^{-4\pi i n l / N_t} \left(\tilde{\varphi}[l - N/2] + \tilde{\varphi}[l + N/2] \right), & m = N_f, \end{cases} \quad (\text{A1})$$

The closed-form expressions below are written for $d = 1$ to keep the algebra explicit. We do remark, however, that the orthonormality and normalisation results extend to arbitrary d using the incomplete-beta identity $\nu_d(1-y) + \nu_d(y) = 1$. We restrict to $d = 1$ here purely for pedagogical clarity:

$$\tilde{\varphi}[l] = \sqrt{\frac{2}{N_t}} \begin{cases} 1 & |l_r| < A \\ \cos \left[\frac{\pi}{2} \left(\frac{|l_r| - A}{B} \right) \right] & A \leq |l_r| < A + B \\ 0 & |l_r| \geq A + B \end{cases} \quad (\text{A2})$$

$$l_r = l \frac{2}{N_t}$$

where $l \in \{-\frac{N_t}{2}, -\frac{N_t}{2} + 1, \dots, \frac{N_t}{2} - 1\}$. Now we verify that

$$\begin{aligned} \sum_{nm} \tilde{g}_{nm}[l] \tilde{g}_{nm}^*[l'] &= \frac{1}{2} \sum_n e^{-4\pi i n (l-l')/N_t} \tilde{\varphi}[l] \tilde{\varphi}[l'] \\ &+ \frac{1}{2} \sum_{0 < m < N_f} \sum_n e^{-2\pi i n (l-l')/N_t} \left(C_{nm} \tilde{\varphi}[l - mN_t/2] + C_{nm}^* \tilde{\varphi}[l + mN_t/2] \right) \left(C_{nm}^* \tilde{\varphi}[l' - mN_t/2] + C_{nm} \tilde{\varphi}[l' + mN_t/2] \right) \\ &+ \frac{1}{2} \sum_n e^{-4\pi i n (l-l')/N_t} \left(\tilde{\varphi}[l - N/2] + \tilde{\varphi}[l + N/2] \right) \left(\tilde{\varphi}[l' - N/2] + \tilde{\varphi}[l' + N/2] \right) \end{aligned} \quad (\text{A3})$$

Now we simplify the $0 < m < N_f$ term, by noticing that $C_{nm}^2 = C_{nm}^{*2} = (-1)^m (-1)^n$ and $|C_{nm}|^2 = 1$

$$\begin{aligned} \frac{1}{2} \sum_{0 < m < N_f} \sum_n e^{-2\pi i n (l-l')/N_t} &\left(\tilde{\varphi}[l - mN_t/2] \tilde{\varphi}[l' - mN_t/2] + \tilde{\varphi}[l + mN_t/2] \tilde{\varphi}[l' + mN_t/2] \right. \\ &\left. + (-1)^m (-1)^n \tilde{\varphi}[l - mN_t/2] \tilde{\varphi}[l' + mN_t/2] + (-1)^m (-1)^n \tilde{\varphi}[l + mN_t/2] \tilde{\varphi}[l' - mN_t/2] \right) \end{aligned} \quad (\text{A4})$$

one notices that the summation over n can be directly performed, and it vanishes for every value but $l = l'$. In particular, the second line (the mixed terms) always vanishes. A similar result holds for the $m = 0, N_f$ cases, therefore we can write

$$\sum_{nm} \tilde{g}_{nm}[l] \tilde{g}_{nm}^*[l'] = \delta_{ll'} \frac{N_t}{2} \left(\tilde{\varphi}^2[l] + \sum_{0 < m < N_f} \tilde{\varphi}^2[l - mN_t/2] + \sum_{0 < m < N_f} \tilde{\varphi}^2[l + mN_t/2] + \tilde{\varphi}^2[l - N/2] + \tilde{\varphi}^2[l + N/2] \right) \quad (\text{A5})$$

Now the thing to notice is that each value of l lays in two and only two of these windows. Let be $\bar{m}$ and $\bar{m} + 1$ the indices of these two windows, such that $\frac{\bar{m}N_t}{2} < |l| < \frac{(\bar{m}+1)N_t}{2}$ ($\bar{m}$ could be 0, or $\bar{m} + 1$ can be N_f). Then the sum becomes

$$\sum_{nm} \tilde{g}_{nm}[l] \tilde{g}_{nm}^*[l'] = \delta_{ll'} \frac{N_t}{2} \left(\tilde{\varphi}^2[l - \bar{m}N_t/2] + \tilde{\varphi}^2[l - (\bar{m} + 1)N_t/2] \right) \quad (\text{A6})$$

Let $l \geq 0$ for sake of simplicity ($l < 0$ cases are analogous). Then either $l < \frac{(m+A)N_t}{2}$ or $l \geq \frac{(m+A)N_t}{2}$. In the first case, we have $\tilde{\varphi}^2[l - \bar{m}N_t/2] = \frac{2}{N_t}$ and $\tilde{\varphi}^2[l - (\bar{m} + 1)N_t/2] = 0$, since for $|\frac{2l}{N_t} - (\bar{m} + 1)| > A + B$. In the second

case, we have

$$\sum_{nm} \tilde{g}_{nm}[l] \tilde{g}_{nm}^*[l'] = \delta_{ll'} \left(\cos^2 \left[\frac{\pi}{2} \frac{\frac{2l}{N_t} - \bar{m} - A}{B} \right] + \cos^2 \left[\frac{\pi}{2} \frac{\bar{m} + 1 - \frac{2l}{N_t} - A}{B} \right] \right) \quad (\text{A7})$$

which after some algebra simplifies to

$$\sum_{nm} \tilde{g}_{nm}[l] \tilde{g}_{nm}^*[l'] = \delta_{ll'} \left(1 + \cos \left(\frac{\pi}{2} \frac{1 - 2A}{B} \right) \cos \left(2 \frac{\pi}{2} \frac{\frac{2l}{N_t} - \bar{m} - A}{B} - \frac{\pi}{2} \frac{1 - 2A}{B} \right) \right) = \delta_{ll'} \quad (\text{A8})$$

where we used the fact of $2A + B = 1$. Therefore we find that for every value of l we have

$$\sum_{nm} \tilde{g}_{nm}[l] \tilde{g}_{nm}^*[l'] = \delta_{ll'} \quad (\text{A9})$$

1. The second (quasi)normality condition

We want to evaluate

$$\sum_l \tilde{g}_{nm}[l] \tilde{g}_{pq}^*[l] \quad (\text{A10})$$

we immediatly note that, since the window support in frequency is $N_t/2$ we are forced to have $q = m$. Now we have to consider three cases.

Case $m = 0$, for which we have

$$\begin{aligned} \sum_l \tilde{g}_{n0}[l] \tilde{g}_{p0}^*[l] &= \frac{1}{2} \sum_{l=-N_t/2}^{N_t/2-1} e^{-4\pi i l(n-p)/N_t} \tilde{\varphi}^2[l] = \frac{1}{2} \sum_{l=-N_t/2}^{N_t/2-1} e^{-4\pi i l(n-p)/N_t} \tilde{\varphi}^2[l] = \\ &= \sum_{l=-N_t/2}^{-1} \left(e^{-4\pi i l(n-p)/N_t} \tilde{\varphi}^2[l] + e^{-4\pi i (l+N_t/2)(n-p)/N_t} \tilde{\varphi}^2[l + N_t/2] \right) \end{aligned} \quad (\text{A11})$$

but now we note that $e^{-4\pi i (l+N_t/2)(n-p)/N_t} = e^{-4\pi i l(n-p)/N_t}$ and in the previous subsection of this appendix we have also shown that for $|l| \leq N_t/2$ we have $\tilde{\varphi}^2[l] + \tilde{\varphi}^2[l + N_t/2] = \frac{2}{N_t}$, therefore the last expression simplifies to

$$\sum_l \tilde{g}_{n0}[l] \tilde{g}_{p0}^*[l] = \frac{1}{N_t} \sum_{l=-N_t/2}^{-1} e^{-4\pi i l(n-p)/N_t} \quad (\text{A12})$$

this is a geometric series in l , which vanishes for every choice of $n - m$, but the only two cases $n - m = 0$ and $n - m = \pm N_t/2$, for which the exponential evaluates to 1, therefore obtaining, for the first case

$$\sum_l \tilde{g}_{n0}[l] \tilde{g}_{p0}^*[l] = \frac{1}{2} \left(\delta_{n,p} + \delta_{n,p \pm \frac{N_t}{2}} \right) \quad (\text{A13})$$

Case $m = N_f$, for which we have

$$\begin{aligned}
\sum_l \tilde{g}_{nN_f}[l] \tilde{g}_{pN_f}^*[l] &= \frac{1}{2} \sum_{l=-N/2}^{N/2-1} e^{-4\pi i l(n-p)/N_t} (\tilde{\varphi}[l - N/2] + \tilde{\varphi}[l + N/2])^2 = \\
&= \frac{1}{2} \sum_{l=-N/2}^{N/2-1} e^{-4\pi i l(n-p)/N_t} (\tilde{\varphi}^2[l - N/2] + \tilde{\varphi}^2[l + N/2]) = \\
&= \frac{1}{2} \sum_{l=-N}^{-1} e^{-4\pi i (l+N/2)(n-p)/N_t} \tilde{\varphi}^2[l] + \frac{1}{2} \sum_{l=0}^{N-1} e^{-4\pi i (l-N/2)(n-p)/N_t} \tilde{\varphi}^2[l] = \\
&= \frac{1}{2} \sum_{l=-N_t/2}^{-1} e^{-4\pi i l(n-p)/N_t} \tilde{\varphi}^2[l] + \frac{1}{2} \sum_{l=0}^{N_t/2-1} e^{-4\pi i l(n-p)/N_t} \tilde{\varphi}^2[l] = \\
&= \frac{1}{2} \sum_{l=-N_t/2}^{N_t/2-1} e^{-4\pi i l(n-p)/N_t} \tilde{\varphi}^2[l]
\end{aligned} \tag{A14}$$

and from here the proof is identical to the $m = 0$ case, leading to

$$\sum_l \tilde{g}_{nN_f}[l] \tilde{g}_{pN_f}^*[l] = \frac{1}{2} \left(\delta_{n,p} + \delta_{n,p \pm \frac{N_t}{2}} \right) \tag{A15}$$

Case $0 < m < N_f$, for which we have

$$\sum_l \tilde{g}_{nm}[l] \tilde{g}_{pm}^*[l] = \frac{1}{2} \sum_{l=-N/2}^{N/2-1} \left[e^{-2\pi i l(n-p)/N_t} \left(C_{nm} \tilde{\varphi}[l - mN_t/2] + C_{nm}^* \tilde{\varphi}[l + mN_t/2] \right) \right. \tag{A16}$$

$$\begin{aligned}
&\times \left(C_{pm}^* \tilde{\varphi}[l - mN_t/2] + C_{pm} \tilde{\varphi}[l + mN_t/2] \right) \Big] \\
&= \frac{1}{2} \sum_{l=-N/2}^{N/2-1} e^{-2\pi i l(n-p)/N_t} \left(C_{nm} C_{pm}^* \tilde{\varphi}^2[l - mN_t/2] + C_{nm}^* C_{pm} \tilde{\varphi}^2[l + mN_t/2] \right)
\end{aligned} \tag{A17}$$

Let's take the first term and shift the frequency, rewriting

$$\frac{1}{2} C_{nm} C_{pm}^* \sum_{l=-N_t/2}^{N_t/2-1} e^{-2\pi i (l+mN_t/2)(n-p)/N_t} \tilde{\varphi}^2[l] = \frac{1}{2} C_{nm} C_{pm}^* e^{-\pi i m(n-p)} \sum_{l=-N_t/2}^{N_t/2-1} e^{-2\pi i l(n-p)/N_t} \tilde{\varphi}^2[l] \tag{A18}$$

and also in this case, from here the proof is identical to the $m = 0$ case, but where in this case, given the different exponent, the series is nonvanishing for $n = p$ and not for $n = p \pm N_t/2$. The expression then simplifies to

$$\frac{1}{2} C_{nm} C_{pm}^* \sum_{l=-N_t/2}^{N_t/2-1} e^{-2\pi i (l+mN_t/2)(n-p)/N_t} \tilde{\varphi}^2[l] = \frac{1}{2} C_{nm} C_{nm}^* = \frac{1}{2} \tag{A19}$$

an analogous proof holds for the term with $+mN_t/2$, therefore the full result is that in this case, for $0 < m < N_f$, we find

$$\sum_l \tilde{g}_{nm}[l] \tilde{g}_{pm}^*[l] = \delta_{n,p} \tag{A20}$$

To wrap up all the cases we find

$$\sum_l \tilde{g}_{nm}[l] \tilde{g}_{pq}^*[l] = \delta_{m,q} \begin{cases} \frac{1}{2} \left(\delta_{n,p} + \delta_{n,p \pm \frac{N_t}{2}} \right) & m = \{0, N_f\} \\ \delta_{n,p} & 0 < m < N_f \end{cases} \tag{A21}$$

2. Normalization of the Meyer window

We verify that the Meyer window defined in Eq. (A2) (the discrete case of (??)) is correctly normalized. We compute

$$\begin{aligned} \sum_{l=-N/2}^{N/2-1} \tilde{\varphi}^2[l] &= 2 \sum_{l=0}^{A \frac{N_t}{2}-1} \frac{2}{N_t} - \frac{2}{N_t} + 2 \sum_{l=A \frac{N_t}{2}}^{(A+B) \frac{N_t}{2}-1} \frac{2}{N_t} \cos^2 \left[\frac{\pi}{2} \frac{2l - A}{B} \right] = 2A - \frac{2}{N_t} + \frac{4}{N_t} \sum_{k=0}^{B \frac{N_t}{2}-1} \cos^2 \left[\frac{\pi}{2} \frac{2k}{BN_t} \right] = \\ &= 2A - \frac{2}{N_t} + \frac{4}{N_t} \sum_{k=0}^{B \frac{N_t}{2}-1} \frac{1}{2} + \frac{4}{N_t} \sum_{k=0}^{B \frac{N_t}{2}-1} \frac{1}{2} \cos \left(\frac{2\pi k}{BN_t} \right) = 2A - \frac{2}{N_t} + B + \frac{2}{N_t} = 1 \end{aligned} \quad (\text{A22})$$

where on the first line the term $-\frac{2}{N_t}$ avoids double counting $l = 0$, where the last summation equals $+\frac{2}{N_t}$ after performing the geometrical series and where we remember that $2A + B = 1$ is a property of the window function.

Appendix B: Detailed derivation for the forward and inverse transforms

This appendix provides the full algebraic steps leading to the compact expressions for the forward WDM coefficients presented in Section II. We work in the Fourier domain throughout, writing the forward transform as

$$w_{nm} = \sum_{l=-N/2}^{N/2-1} \tilde{x}_l \tilde{g}_{nm}^*(f[l]), \quad (\text{B1})$$

where the discrete frequencies are mapped as $f[l] = l \Delta f = l/T$.

1. Interior channels ($0 < m < N_f$)

Inserting the interior-channel basis element from Eq. (11) and splitting the sum over positive and negative frequencies gives

$$\begin{aligned} w_{nm} &= \frac{1}{\sqrt{2}} \sum_{l=0}^{N/2-1} \tilde{x}_l e^{2\pi i l n / N_t} \left[C_{nm}^* \tilde{\varphi} \left(\left(l - \frac{mN_t}{2} \right) \Delta f \right) + C_{nm} \tilde{\varphi} \left(\left(l + \frac{mN_t}{2} \right) \Delta f \right) \right] \\ &+ \frac{1}{\sqrt{2}} \sum_{l=-N/2}^{-1} \tilde{x}_l e^{2\pi i l n / N_t} \left[C_{nm}^* \tilde{\varphi} \left(\left(l - \frac{mN_t}{2} \right) \Delta f \right) + C_{nm} \tilde{\varphi} \left(\left(l + \frac{mN_t}{2} \right) \Delta f \right) \right]. \end{aligned} \quad (\text{B2})$$

Using the reality condition $\tilde{x}_{-l} = \tilde{x}_l^*$ together with the symmetry $\tilde{\varphi}(-f) = \tilde{\varphi}^*(f)$, the second sum is the complex conjugate of the first. The boundary terms at $l = 0$ and $l = -N/2$ vanish because the window support does not extend to those frequencies for interior channels. We therefore obtain

$$\begin{aligned} w_{nm} &= \sqrt{2} \Re \sum_{l=1}^{N/2-1} \tilde{x}_l e^{2\pi i l n / N_t} C_{nm}^* \tilde{\varphi} \left(\left(l - \frac{mN_t}{2} \right) \Delta f \right) \\ &= \sqrt{2} (-1)^{nm} \Re \sum_{l'=-N_t/2}^{N_t/2} \tilde{x}_{l'+mN_t/2} e^{2\pi i l' n / N_t} C_{nm}^* \tilde{\varphi}(l' \Delta f), \end{aligned} \quad (\text{B3})$$

where the last equality follows from the substitution $l' = l - mN_t/2$ and the fact that $\tilde{\varphi}(l' \Delta f)$ is nonzero only for $|l'| \leq N_t/2$. This is Eq. (17) of the main text, and corresponds (up to normalization conventions) to Eq. 17 of Cornish [5].

2. DC channel ($m = 0$)

For the DC edge channel, the basis element has a doubled time-shift exponent and no C_{nm} factor. The same positive/negative frequency splitting yields

$$\begin{aligned} w_{n0} &= \frac{1}{\sqrt{2}} \sum_{l=0}^{N/2-1} \tilde{x}_l e^{4\pi i l n / N_t} \tilde{\varphi}(l \Delta f) + \frac{1}{\sqrt{2}} \sum_{l=-N/2}^{-1} \tilde{x}_l e^{4\pi i l n / N_t} \tilde{\varphi}(l \Delta f) \\ &= \sqrt{2} \Re \left[\sum_{l=1}^{N_t/2} \tilde{x}_l e^{4\pi i l n / N_t} \tilde{\varphi}(l \Delta f) \right] + \frac{1}{\sqrt{2}} \tilde{x}_0 \tilde{\varphi}(0), \end{aligned} \quad (\text{B4})$$

which is Eq. (20) of the main text.

3. Nyquist channel ($m = N_f$)

$$\begin{aligned} w_{nm} &= \frac{1}{\sqrt{2}} \sum_{l=0}^{N/2-1} \tilde{x}_l e^{2\pi i l n / N_t} [\tilde{\varphi}((l - \frac{N}{2})\Delta f) + \tilde{\varphi}((l + \frac{N}{2})\Delta f)] + \frac{1}{\sqrt{2}} \sum_{l=-N/2}^{-1} \tilde{x}_l e^{2\pi i l n / N_t} [\tilde{\varphi}((l - \frac{N}{2})\Delta f) + \tilde{\varphi}((l + \frac{N}{2})\Delta f)] \\ &= \frac{1}{\sqrt{2}} \sum_{l=0}^{N/2-1} \tilde{x}_l e^{2\pi i l n / N_t} [\tilde{\varphi}((l - \frac{N}{2})\Delta f) + \tilde{\varphi}((l + \frac{N}{2})\Delta f)] + \frac{1}{\sqrt{2}} \sum_{l=1}^{N/2} \tilde{x}_{-l} e^{-2\pi i l n / N_t} [\tilde{\varphi}((-l - \frac{N}{2})\Delta f) + \tilde{\varphi}((-l + \frac{N}{2})\Delta f)]. \end{aligned} \quad (\text{B5})$$

the $l = 0$ term in the first summation vanishes, and the $l = N/2$ term is treated separately:

$$\begin{aligned} w_{nm} &= \sqrt{2} \Re \sum_{l=1}^{N/2-1} \tilde{x}_l e^{2\pi i l n / N_t} \tilde{\varphi}((l - \frac{N}{2})\Delta f) + \frac{1}{\sqrt{2}} \tilde{x}_{N/2} e^{-\pi i n N_f} \tilde{\varphi}(0) \\ &= \sqrt{2} \Re \sum_{l'=-N_t/2}^{-1} \tilde{x}_{l'+N/2} e^{2\pi i l' n / N_t} \tilde{\varphi}(l' \Delta f) + \frac{1}{\sqrt{2}} \tilde{x}_{N/2} \tilde{\varphi}(0), \end{aligned} \quad (\text{B6})$$

assuming an even value for N_f .

4. The inverse transform

We write the inverse transform of the w_{nm} coefficients to recover the signal $x[l]$ for $l \geq 0$ since we know that for $l < 0$ the reality condition holds ($x[-l] = x^*[l]$):

$$\begin{aligned} x[l] &= \sum_{n=0}^{N_t-1} \sum_{m=0}^{N_f} w_{nm} g_{nm}[l] = \frac{1}{\sqrt{2}} \sum_n \sum_{0 < m < N_f} w_{nm} e^{-2\pi i l n / N_t} [C_{nm} \tilde{\varphi}((l - \frac{m N_t}{2})\Delta f) + C_{nm}^* \tilde{\varphi}((l + \frac{m N_t}{2})\Delta f)] \\ &\quad + \frac{1}{\sqrt{2}} \sum_n w_{n0} e^{-4\pi i l n / N_t} \tilde{\varphi}(l \Delta f) + \frac{1}{\sqrt{2}} \sum_n w_{nN_f} e^{-4\pi i l n / N_t} [\tilde{\varphi}((l - \frac{N}{2})\Delta f) + \tilde{\varphi}((l + \frac{N}{2})\Delta f)] = \\ &= \begin{cases} x_0[l] & l < N_t/2 \\ x_m[l] & m N_t/2 \leq l < (m+1) N_t/2 \\ x_{N_f}[l] & l \geq N/2 - N_t/2 \end{cases} \end{aligned} \quad (\text{B7})$$

where we note that for each value of l , the inverse transform depends on which of the m bands the l value belongs, in particular

$$\begin{aligned} x_0[l] &= \frac{1}{\sqrt{2}} \sum_n w_{n1} e^{-2\pi i l n / N_t} C_{n1} \tilde{\varphi}\left(\left(l - \frac{N_t}{2}\right) \Delta f\right) + \frac{1}{\sqrt{2}} \sum_n w_{n0} e^{-4\pi i l n / N_t} \tilde{\varphi}(l \Delta f) \\ x_m[l] &= \frac{1}{\sqrt{2}} \sum_n w_{nm} e^{-2\pi i l n / N_t} C_{nm} \tilde{\varphi}\left(\left(l - \frac{m N_t}{2}\right) \Delta f\right) + \frac{1}{\sqrt{2}} \sum_n w_{n(m+1)} e^{-2\pi i l n / N_t} C_{n(m+1)} \tilde{\varphi}\left(\left(l - \frac{(m+1) N_t}{2}\right) \Delta f\right) \\ x_{N_f}[l] &= \frac{1}{\sqrt{2}} \sum_n w_{n(N_f-1)} e^{-2\pi i l n / N_t} C_{n(N_f-1)} \tilde{\varphi}\left(\left(l - \frac{N}{2} + \frac{N_t}{2}\right) \Delta f\right) + \frac{1}{\sqrt{2}} \sum_n w_{n N_f} e^{-4\pi i l n / N_t} \tilde{\varphi}\left(\left(l - \frac{N}{2}\right) \Delta f\right) \end{aligned} \quad (\text{B8})$$

Which can be written in a slightly more convenient form:

$$\tilde{x}_m[l] = \frac{1}{\sqrt{2}} \begin{cases} \tilde{\varphi}[l] \sum_{n=0}^{N_t-1} w_{n0} e^{-4\pi i n l / N_t}, & m = 0, \quad 0 \leq l < N_t/2, \\ \tilde{\varphi}[l - m N_t/2] \sum_{n=0}^{N_t-1} C_{nm} w_{nm} e^{-2\pi i n l / N_t}, & 1 \leq m \leq N_f - 1, \quad (m-1) \frac{N_t}{2} \leq l < (m+1) \frac{N_t}{2}, \\ \tilde{\varphi}[l - N/2] \sum_{n=0}^{N_t-1} w_{n N_f} e^{-4\pi i n l / N_t}, & m = N_f, \quad N/2 - N_t/2 \leq l \leq N/2, \end{cases} \quad (\text{B9})$$

which is presented in the main text (25).

Appendix C: The Local Stationary Assumption

1. The Evolutionary PSD

Let us define the discrete time-domain auto-covariance function for a stochastic (real) process x via

$$R[k_1, k_2] = \mathbb{E}[n[k_1], n[k_2]] \quad (\text{C1})$$

for time-bins $k_1, k_2 \in \{0, \dots, N-1\}$. For the general auto-covariance function (C1), the stationary assertion means that

$$R[k_1, k_2] = R[0, k_1 - k_2] = \mathbb{E}[n[0] n[k_1 - k_2]] = R(k) \quad (\text{C2})$$

for $k = k_1 - k_2$ a lag parameter. Such a condition forces the time-domain noise covariance matrix to obey a Toeplitz like structure (not necessarily circulant). In this section, however, we relax the assumption that the noise process is stationary and instead consider a general Gaussian stochastic process x that may not be stationary. Define $\bar{k} = (k_1 + k_2)/2$ and $\kappa = k_1 - k_2$, then the auto-covariance function takes the form

$$R(\bar{k} + \kappa/2, \bar{k} - \kappa/2) \quad (\text{C3})$$

At this point, we can now define the discrete evolutionary PSD via the Fourier transform of the (non-toeplitz) time-domain auto covariance function

$$S[\bar{k}, l] = 2\Delta t \sum_{\kappa=0}^{N-1} R(\bar{k} + \kappa/2, \bar{k} - \kappa/2) \exp(-2\pi i l \kappa / N) \quad (\text{C4})$$

Notice that this reduces to the usual definition of the Power Spectral Density if the parameter $\bar{k}$ (asserting time-dependence) is dropped from the PSD. The factor of 2 results from using a one-sided PSD and Δt from the sum.

It is then possible to build the frequency domain noise covariance matrix via

$$\mathbb{E}[\tilde{x}[l], \tilde{x}^*[l']] = \sum_{k_1, k_2} \mathbb{E}[k_1 k_2] \exp(-2\pi i(k_2 l' - k_1 l)/N) \quad (\text{C5})$$

$$= \frac{1}{2\Delta t} \sum_{\bar{k}} \left\{ 2\Delta t \sum_{\kappa} R[\bar{k} + \kappa/2, \bar{k} - \kappa/2] \exp\left(-2\pi i \left[\frac{l+l'}{2}\right] \kappa/N\right) \right\} \exp(2\pi i \bar{k}(l-l')/N) \quad (\text{C6})$$

$$= \frac{1}{2\Delta t} \sum_{\bar{k}} S\left[\bar{k}, \frac{1}{2}(l+l')\right] \exp(-2\pi i \bar{k}(l-l')/N), \quad (\text{C7})$$

also known as the Wigner-Ville representation [Ollie: cite]. Notice for arbitrary values of κ the frequency domain noise covariance matrix now has non-trivial off diagonal elements that must be accounted for during inference. Notice that if we drop $\bar{k}$ from the evolutionary PSD and evaluating the sum over the complex exponential, we obtain the familiar stationary expression

$$\mathbb{E}[\tilde{x}[l], \tilde{x}^*[l']] = \frac{NS[l]}{2\Delta t} \delta_{ll'}. \quad (\text{C8})$$

[Ollie: Careful Ollie – Toeplitz + Circulant. I think what we need to say here is that R encodes all of the statistical information about the process x , therefore being circulant (periodic statistics at end points).]

2. The WDM Covariance Matrix

Consider now the expression for the WDM covariance matrix

$$(\Sigma_w)_{(n,m),(p,q)} = \sum_{l,l'} \tilde{g}_{nm}^*[l] \tilde{g}_{pq}[l'] \mathbb{E}[\tilde{x}^*[l] \tilde{x}[l']] \quad (\text{C9})$$

$$= \frac{1}{2\Delta t} \sum_{l,l',\bar{k}} \tilde{g}_{nm}^*[l] \tilde{g}_{pq}[l'] \sum_{\bar{k}} S\left[\bar{k}, \frac{1}{2}(l+l')\right] \exp(-2\pi i \bar{k}(l-l')/N), \quad (\text{C10})$$

We will now look at two separate cases – the leading diagonal $(n,m) = (p,q)$ and off diagonal elements $(n,m) \neq (p,q)$. The atoms from the Wilson basis $\tilde{g}$ are peaked around the value of $l \approx mN_t/2$. This is where the support of the Wilson basis lies. This means that the dominant contribution of the sum are for those l, l' in the vicinity of $mN_t/2$ and $qN_t/2$. Let us now assume that the evolutionary PSD varies slowly over the regime $(l+l')/2 \approx mN_t/2$ and the temporal time index $\bar{k} \approx nN_f$

Therefore, let us expand $S[\bar{k}, (l+l')/2]$ in this vicinity and applying the local stationary approximation

$$S\left[\bar{k}, \frac{1}{2}(l+l')\right] \approx S[nN_f, mN_t/2] + \left(\frac{l+l'}{2} - mN_t/2\right) \partial_f S + (\bar{k} - nN_f) \partial_t S + \text{corrections} \quad (\text{C11})$$

Now, for the leading diagonal, substituting the leading order term from (C11) into (C10) and evaluating the sum, one obtains

$$(\Sigma_w)_{(n,m),(p,q)} \approx \frac{1}{2\Delta t} S[nN_f, mN_t/2] \sum_{l,l',\bar{k}} \tilde{g}_{nm}^*[l] \tilde{g}_{nm}[l'] \exp(-2\pi i \bar{k}(l-l')/N), \quad (\text{C12})$$

$$= \frac{N}{2\Delta t} S[nN_f, mN_t/2] \sum_l |\tilde{g}_{nm}[l]|^2 \quad (\text{C13})$$

$$= \frac{N}{2\Delta t} S[n, mN_t/2]. \quad (\text{C14})$$

Now let's look at each part of the first order contributions in the evolutionary PSD

$$\frac{1}{2\Delta t} \partial_f S[nN_f, mN_t/2] \sum_{l, l', \bar{k}} \tilde{g}_{nm}^*[l] \tilde{g}_{nm}[l'] \left(\frac{l + l'}{2} - mN_t/2 \right) \exp(-2\pi i \bar{k}(l - l')/N), \quad (C15)$$

$$= \frac{N}{2\Delta t} \partial_f S[nN_f, mN_t/2] \sum_l |\tilde{g}_{nm}[l]|^2 [l - mN_t/2], \quad (C16)$$

$$= 0? \quad (C17)$$

[Ollie: I think this is zero because of the Meyer window properties. Can you check Giorgio?] The support of $\tilde{g}$ is in the vicinity of $l = mN_t/2$. Loosely speaking, when $\tilde{g}$ is maximized, it is then suppressed by the factor $l - mN_t/2$. The temporal term takes the form:

$$\frac{1}{2\Delta t} \partial_t S[nN_f, mN_t/2] \sum_{l, l', \bar{k}} \tilde{g}_{nm}^*[l] \tilde{g}_{nm}[l'] (\bar{k} - nN_f) \exp(-2\pi i \bar{k}(l - l')/N), \quad (C18)$$

$$= \frac{1}{2\Delta t} \partial_t S[nN_f, mN_t/2] \sum_{\bar{k}} \left[\left(\sum_l \tilde{g}_{nm}^* \exp(2\pi i \bar{k}l/N) \sum_{l'} \tilde{g}_{nm}[l'] \exp(-2\pi i l'/N) \right) (\bar{k} - nN_f) \right] \quad (C19)$$

$$= \frac{N^2}{2\Delta t} \partial_t S[nN_f, mN_t/2] \sum_{\bar{k}} |g_{nm}[\bar{k}]|^2 (\bar{k} - nN_f), \quad (C20)$$

Where we have identified inverse Fourier transforms of the $\tilde{g}$ basis functions to dramatically simplify the expression above.

From Necula [21], the time-domain Wilson basis follows from inverse-DFT of Eq. (11):

$$g_{nm}[\bar{k}] = \sqrt{2} \varphi[\bar{k} - nN_f] \cos\left(\frac{\pi m \bar{k}}{N_f} - \theta_{nm}\right), \quad (C21)$$

for the alternating phase factor θ_{nm}

$$\theta_{nm} = \begin{cases} 0 & (n + m) \text{ even} \\ \pi/2 & (n + m) \text{ odd} \end{cases} \quad (C22)$$

where φ is the inverse DFT of the Meyer window $\tilde{\varphi}$ and θ_{nm} . The atom is localised in time at the centre $k = nN_f$ with width set by the inverse-DFT of $\tilde{\varphi}$, and modulated at the channel frequency $m\Delta F$.

[Ollie: Loose notes –] Now, the $\cos^2 \sim \frac{1}{2} + \text{oscillatory}$, so, perhaps, I feel like the quantity

$$\sum_{\bar{k}} |g_{nm}[\bar{k}]|^2 (\bar{k} - nN_f) \approx \frac{1}{\sqrt{2}} \sum_{\bar{k}} (\bar{k} - nN_f) |\varphi[\bar{k}]|^2 \approx 0 \quad (C23)$$

since φ is an even function $\varphi[-k] = \varphi[k]$, around its centre nN_f . This implies that the temporal piece of the evolutionary power spectral density also vanishes. The diagonal components therefore take the form

$$(\Sigma_w)_{(n,m),(n,m)} = \frac{N}{2\Delta t} S[nN_f, mN_t/2] [1 + \mathcal{O}(\text{quadratic corrections})] \quad (C24)$$

in other words, in the locally stationary limit. We leave the off-diagonal elements to another day.

In the very simple case of wide sense stationary noise, $\partial_t S = 0$, the pre-factor (involving sums) of $\partial_f S$ is still zero. It can then be shown that

$$(\Sigma_w)_{(n,m),(p,q)} \approx \frac{1}{2\Delta t} S[nN_f, mN_t/2] \sum_{l, l', \bar{k}} \tilde{g}_{nm}^*[l] \tilde{g}_{pq}[l'] \exp(-2\pi i \bar{k}(l - l')/N), \quad (C25)$$

$$= \frac{N}{2\Delta t} S[nN_f, mN_t/2] \sum_l \tilde{g}_{nm}^*[l] \tilde{g}_{pq}[l] \quad (C26)$$

$$= \frac{N}{2\Delta t} S[nN_f, mN_t/2] \delta_{np} \delta_{mq} \quad (C27)$$

$$= \delta_{np} \delta_{mq} S_{nm}^{\text{WDM}}. \quad (C28)$$

with quantity

$$S_{nm}^{\text{WDM}} = \frac{N}{2\Delta t} S[nN_f, mN_t/2] \quad (\text{C29})$$

the WDM-based evolutionary power spectral density, evaluated at $t_n = (nN_f)\Delta T$ and $f_m = (mN_t/2)\Delta F$.

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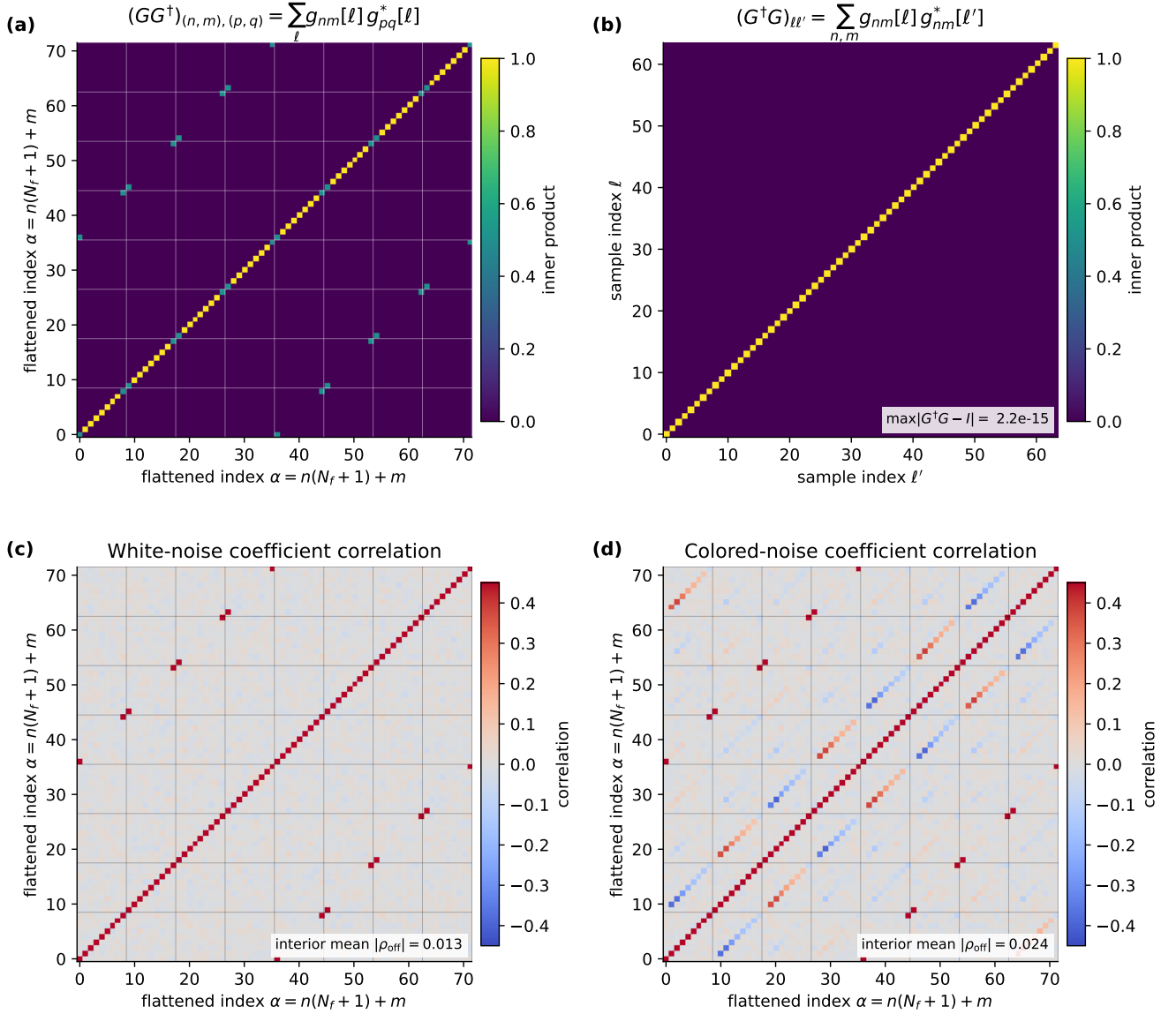


Figure 2. **[Giorgio: Ollie was suggesting (and I would also agree) to replace this figure with the $\sum_{\ell} g_{nm}[\ell] g_{pq}^*[\ell]$ and $\sum_{n,m} g_{nm}[\ell] g_{nm}^*[\ell']$ plots. I can recommend setting $N = 64, N_t = 8, N_f = 8$ and flattening the wdm array: what do you think about it, Avi?] Numerical checks of orthogonality and decorrelation.** Left: Gram matrix of the interior sampled WDM atoms, showing the discrete inner products $\langle g_i, g_j \rangle$ after flattening the interior index pair (n, m) into a single label i . The packed-storage atoms used here are mutually orthogonal but not unit-normalised, so the diagonal of G does not equal 1; the relevant numerical check is that the off-diagonal entries vanish at machine precision ($\max_{i \neq j} |G_{ij}| \sim 10^{-14}$), providing a direct demonstration of orthogonality for the interior channels. Middle: empirical correlation matrix of interior WDM coefficients estimated from many realizations of white Gaussian noise; most off-diagonal entries remain small. Right: the same calculation for colored stationary noise with a 4th-order Butterworth-type PSD $S(f) \propto [1 + (f/f_c)^4]^{-1}$, $f_c = 0.06$ Hz (unit sampling rate), where visibly stronger off-diagonal correlations appear, in agreement with [7], where the correlations between (n, m) and (n', m') appear only at the closest two values of $m' = 0, \pm 1$ and decaying with $|n - n'|$. A total number of samples $N = 64$ is assumed, and the WDM transform is obtained with $N_t = N_f = 8$.

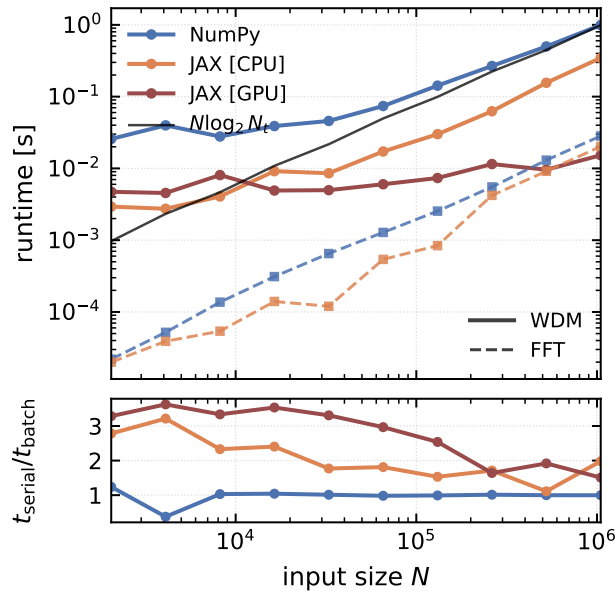


Figure 3. **Forward WDM runtime versus input length.** Top: median wall-clock runtime over 7 runs after one warmup call; shaded bands span the 16th–84th percentile interval. The black dashed line shows the $N \log_2 N$ reference scaling. Bottom: batched-execution speedup $t_{\text{serial}}/t_{\text{batch}}$, where the serial baseline applies the kernel independently to each of three channels and the batched call processes all three simultaneously. Reported timings exclude JAX JIT compilation and host–device data transfer. [Lore: I think that calling the y-axis of the bottom plot just "speed-up" would make it easier for the reader] [Avi: sure, can do. Also i will plot $n\log(nt)$ and FFT timing, up to $N = 10^7$.]



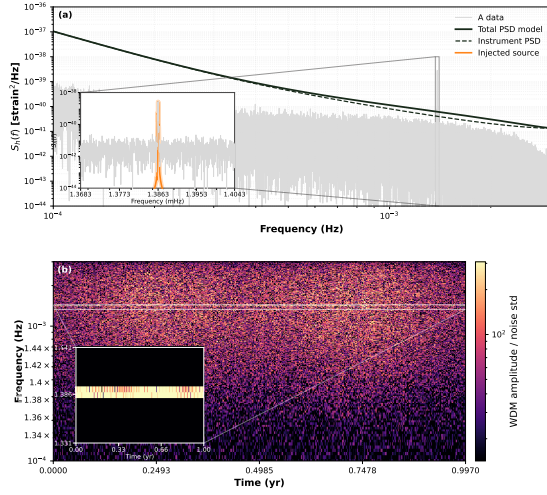


Figure 4. **LISA galactic background: Strain comparison.** Characteristic-strain comparison between the injected resolved galactic binary and the channel noise model including the galactic foreground. **[Giorgio: Quentin says it's a bit confusing to have A data and PSD in the same plot (A data not following the total PSD curve because it is not a PSD, but $|h|^2$). Shall we renormalize the data to match the PSD?]**

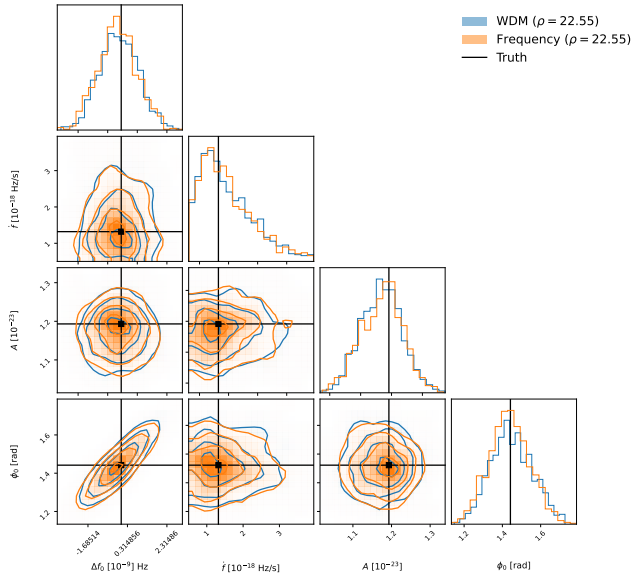


Figure 5. **LISA galactic background: Posterior analysis.** Overlaid posterior corner plot for the WDM-domain and frequency-domain analyses of the same realization. The two posterior families overlap closely in $(f_0, \dot{f}, A, \phi_0)$ and in the derived optimal SNR.

